

**VIETNAM NATIONAL UNIVERSITY – HOCHIMINH CITY
INTERNATIONAL UNIVERSITY**



SCHOOL OF BIOTECHNOLOGY

Guidelines for BSc Thesis Preparation



For internal use only

Updated June, 2021

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PART 1: MUST-KNOW ISSUES

1. Enrolment and Pre-requisites

Your research project begins in your last semester. The project is considered as a 12-credit course which must be completed within the same semester to qualify for graduation. Other important courses such as Biostatistics, Scientific Writing Workshop and Research Methodology should be taken prior to the start of your thesis project.

2. Goals and Objectives

The aim of the research project is to provide students with practice on how to undertake an original research in the major fields offered by the International University (IU)-School of Biotechnology (SBT): Biotechnology, Food Technology and Aquatic Resource Management. The results will be presented to an Evaluation Committee set up by the school and authorized by the Rector of IU. By the end of the research project students will have gained experience in conducting an independent research and should be capable in it.

3. Duration and workload

The research project comprises a 12-credit module equivalent to four working months. The official starting date of thesis project is indicated in a decision paper issued by the Rector's Office. Students are expected to devote regular time in preparing the research proposal, commencing the research project, writing the thesis and presenting it before an Evaluation Committee.

4. Scope

Projects should be original laboratory, field-based or survey research on a topic proposed by your adviser at IU. You could also conduct their thesis project outside the University given that your proposal is approved with adequate supervision by external supervisor.

5. Choice of projects

SBT and its faculty members will offer a list of possible projects for students' consideration. The proposed projects are closely related to the supervisor's expertise and considered feasible given the current conditions of the IU laboratory system or alternatives elsewhere. Students can select the project they are most interested in and discuss with the faculty member proposing the project. Competition may exist when more than one student is

interested in the same project. The supervisor has the right to select the most suitable student but criteria for selection should be publicized.

It is possible for students to propose and arrange these projects themselves, but the topic and scientific content must be endorsed by an Evaluation Committee of the School for final approval by the Rector of the International University. For project that will be conducted outside IU and supervised by non-IU employer, students are requested to provide evidence for such an arrangement by completing Form BT01 along with a CV of your supervisor.

6. Assessment

The thesis will be evaluated by an anonymous reviewer assigned by the School. Students are allowed to present his/her thesis only if the mark is higher than 50/100. Your supervisor is requested to mark your performance, using Form BT03 and send the result to the School. The final mark is the average given by five members of the Scientific Committee, the reviewer and your supervisor.

8. Adviser-Student Relationship

The goal of both student and academic supervisor including the Evaluation Committee is towards the successful completion of the project. Everyone must cooperate and provide full support in the achievement of this goal. Students are required to meet advisor weekly to discuss the research progress. This is to ensure that students follow strictly the work plan approved the adviser. Also, this allows the adviser to assist student in dealing with problems rapidly before they escalate into bigger problems. Should problems emerge during the course of the study, they should be referred immediately to the School via the Academic Affairs Secretaries.

9. Progress report and change of research topic

- About eight weeks after the start of your research you are required to submit a progress report to the school using Form BT02. This progress report must be certified by the supervisor. Change of the initial research title and/or objectives, if well justified, are possible and should be officially approved by the School.
- Students who wish to extend the research duration should indicate their request in the Progress Report or in a separate request form and seek for the School's approval. However, any delay in finishing the research project could possibly result in a significant set back in thesis defence and graduation. Therefore, students are strongly urged to maintain their schedule and work hard for the timely completion of their thesis projects.

10. Thesis submission and revision

- The date for submission of completed theses is set by the School (i.e. four months upon commencement of the research) and will be confirmed before the beginning of the semester.
- Two copies of thesis (soft-bounded) should be submitted to the School' secretaries

two weeks before the date set for thesis defence.

- After a successful defence, the student revises his/her thesis according to the comments and amendments required by the Evaluation Committee. The advisor should make sure that all corrections are followed by the student by approving the revised thesis using Form BT07. Students should contact the secretary of the Evaluation Committee to receive a copy of the defence memo with detailed requests of the Committee for revision/correction of thesis.
- The revised thesis is finally checked and approved by the Committee' secretary on the behalf of the School.
- Students are required to submit two copies of thesis (no binding is required) and a CDROM that contains the electronic versions of the thesis (in both .doc and /pdf formats) and the presentation in PowerPoint. The CDROM should be labelled with student name, ID and year of graduation.

PART 2: THESIS CONTENT

From 2012 onwards students are required to write theses in the form of an extended paper. This new requirement is not only to train SBT' students with manuscript preparation, but also to facilitate later publication of good research by the School. For your thesis the following sections are required in the order shown below. Start each section on a new page.

- Cover page: use the format issued by the School
- Acknowledgment
- Main body: paper-styled, including
 - + *Title, author name(s) and affiliation*
 - + *Abstract*
 - + *Introduction*
 - + *Materials and Methods*
 - + *Results*
 - + *Discussion*
 - + *Conclusion*
 - + *References*
- Appendix (if needed only)

ACKNOWLEDGMENT

This section is to recognize the people, and institutions who have helped you in completing your research project. The page is very informal and you can write in any style that you want. It is best to keep this section short. List here those individuals who provided help during the research (e.g., providing funding, language help, writing assistance or proof reading the article, etc.).

ABSTRACT

The abstract is a very brief overview of your entire study. It must come immediately after the title page. The abstract should briefly state the purpose of the research (introduction), how the problem was studied (methods), the important findings (results), and what the findings mean (conclusion). In summary, it is a short, structured (Introduction, Methods, Results and Conclusion), single paragraph summary, from 250 to 500 words. The author should also suggest some keywords that well represent the content of the research.

INTRODUCTION

This section is short (about 2 - 3 pages) and should be comprehensible to an informed lay person and give enough background to enable the reader to place the particular research problem in a context of common knowledge. It is important to state:

- (i) the research problems
- (ii) a snap-shot literature review on what have been known or not known yet in relation to relevant hypotheses or assumptions suggested by you

- (iii) the purposes of your research
- (iv) scope and limitation
- (v) expected outcomes.

More specifically, all problem elements, including the variables to be studied, should be expressed in an orderly system of relationships. Research questions must be clear, consistent, and measurable. They guide the research design process. Indicate "why" the study is being proposed.

Provide an adequate background (literature review) and clearly state the objectives of the work, avoiding a detailed literature survey or a summary of the results. Try to answer the question: "what potential impact will the results of the study have on the current body of knowledge?"

NOTES:

Think of the introduction as a mental road map that must answer for the reader these four questions:

- What was I studying?
- Why was this topic important to investigate?
- What did we know about this topic before I did this study?
- How will this study advance new knowledge or new ways of understanding?

A well-written introduction is important because, quite simply, you never get a second chance to make a good first impression

- The logic of your argument, your writing style, the overall quality of your research, and, ultimately, the validity of your findings and conclusions.
- A vague, disorganized, or error-filled introduction will create a negative impression
- A concise, engaging, and well-written introduction will lead your readers to think highly of your analytical skills, your writing style, and your research approach.

MATERIALS & METHODS

This section should provide an accurate description of all methods and materials used in your study. It should be written in the past tense in the passive voice. Provide sufficient detail to allow the work to be reproduced, with details of supplier and catalogue number when appropriate. Methods already published should be indicated by a reference: only relevant modifications should be described. See Appendix 2 for an example of this section.

Recommended structure of the section:

- 2.1 Research object and location (information about the object of your research and where it was conducted)
- 2.2 Experimental design: describe the experimental design, methods adopted or developed to collect data. Relevant instruments and materials should be mentioned along with their description. Do not just simply list all the chemicals, instruments or devices used in the research. If you use standard methods (published and used by many similar studies, for example Kjeldall method to determine crude protein concentration), just mention the name of the methods and cite the reference that describe the method. In case the method should be described but too long, detailed information can be presented in the Appendix.

- 2.3 Data analysis: describe statistical methods used for data analysis with enough details so that the reliability of your research can be assessed. Data should be analyzed using statistics, either descriptive or inferential or both. Raw data are never included in your thesis unless they are needed to give evidence for specific conclusions which cannot be obtained by looking at an analysis, or summation, of the data.

If your study includes more than one experiment, describe one by one.

RESULTS

Summarize the findings without interpretation. Results should be clear and concise. Only analyzed data should be presented in forms of figures, graphs, tables and/or text descriptions of observations. Avoid using phrases that are vague or non-specific, such as, "appeared to be greater than other variables..." or "demonstrates promising trends that...." Subjective modifiers should be explained in the discussion section of the paper [i.e., why did one variable appear greater? Or, how does the finding demonstrate a promising trend?].

When presenting statistically summarised data, you should state whether the number is a mean or median and clearly state how the data spread is expressed ($\pm$ standard deviation, $\pm$ standard error of the mean, or inter-quartile range). When claiming a statistically significant result, you must support such a statement with a declaration of the probability (p) value and the test that was used to generate that value.

All Figures and Tables should be numbered chronologically as they appear in your thesis. All Figures and Tables must be referred to in the text to facilitate reading. And they should be displayed in embedded data format, not compressed photos/ pictures. See further guidelines for constructing tables and figures in Part 3.

Use the past tense when referring to your results.

DISCUSSION

This should explore the significance of the results of the work, not repeat them. Discuss all the significant outcomes of your research; see how they fit with our current understanding of the research areas or what implications it implies for future studies or industrial application. Any limitation or weakness of the research should also be discussed and ended up with recommendations for possible improvement.

Use the present verb tense, especially for established facts; however, refer to specific works or prior studies in the past tense.

If needed, use subheadings to help organize your discussion or to categorize your interpretations into them

Note:

A combined Results and Discussion section is often appropriate. Avoid extensive citations and discussion of published literature.

CONCLUSION

This section should state the conclusions and recommendations that you have drawn from your work (in relation to the research question or tested hypothesis) and relate the findings of your study to previously published work. Students should avoid to state the key results here instead of conclusions. Recommendations should be relevant to your research findings

in order to provide the readers with tips, suggestions or modes of action so that they can follow if interested.

REFERENCES

This must contain complete list of **all** references cited in the text. Student should use Citation software to make this section uniform (EndNote, Mendeley... are commonly used in scientific writing). (see Section 5.2 on referencing).

APPENDIX

Any other relevant information that cannot be appropriately accommodated elsewhere can be placed in an Appendix (or Appendices) at the end of the dissertation. Try not to use them unless you absolutely have to. They are considered useful for listing raw data or details of experimental protocols if you feel it is necessary to do so.

PART 3: THESIS FORMAT

From 2012 onwards students at the School of Biotechnology are required to write their theses in the form of an extended paper. The format of your thesis is, therefore, a blended design of a traditional thesis, i.e. with the cover page, followed by Acknowledgment and ended up with an Appendix. The main body of the thesis is, however, a paper which is allowed to be a bit longer than the standard. In order to facilitate professional writing the format of Journal of Biotechnology (Elsevier) is adopted. You are advised to strictly follow the instructions below.

THESIS LAYOUT

- ❖ The thesis must be word-processed in English (American or British usage is accepted, but not a mixture of these) using Verdana (font size: 10) or Times New Roman (font size: 12) with 1.5 line spacing. The text should be fully justified and leave 1 space between sentences.
- ❖ Page set-up: use A4 paper with the left margin of 4.0 cm to allow binding. All the other margins are 2.5 cm.
- ❖ Each page of the main body must be numbered, starting with the page that has the title of your research and the abstract. Place the number in the centre of the bottom of the page. No header/footer is allowed.
- ❖ Binding will be arranged by the School once you submit the final version of your thesis.

NUMBER OF PAGES

- ❖ Keep your writing short, informative and as concise as possible.
- ❖ No page number is required for the Cover page, Acknowledgment, References and Appendix.
- ❖ The length of the main body of your thesis should be ideally between 15 and 20 pages. When needed the addition of few more pages are allowed, but the total number of pages of the main body should not exceed 25.
- ❖ Your supervisor will advise you on the length of each section and the level of details required.

COVER PAGE

- ❖ The cover page is designed to highlight your research title while providing important information such as the name of the educational provider, name of student and adviser(s) and year of publication.
- ❖ Use the standard format provided by the School (see Appendix 1).

HEADINGS

The appropriate use of headings is a great assistance to the reader, breaking the text into logical blocks. Divide your thesis into clearly defined and numbered sections. Subsections should be numbered 1.1 (then 1.1.1, 1.1.2, ...), 1.2, etc. Any subsection may be given a brief heading. Each heading should appear on its own separate line. The recommended structure and headings of the main body is as follows:

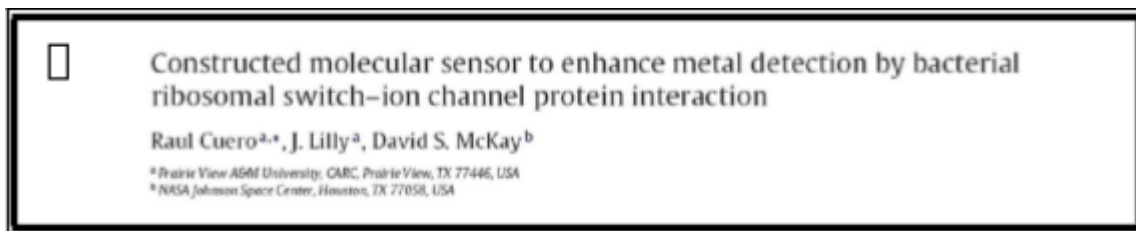
Title

Author name(s) and affiliation

Abstract

Keywords

1. Introduction
 2. Materials & Methods
 - 2.1 Research object and location
 - 2.2 Experimental design
 - 2.3 Data analysis
 3. Results
 - 3.1 sub-headline 1
 - 3.2 sub-headline 2
 - 3.n sub-headline n
 4. Discussion
 5. Conclusion
- References



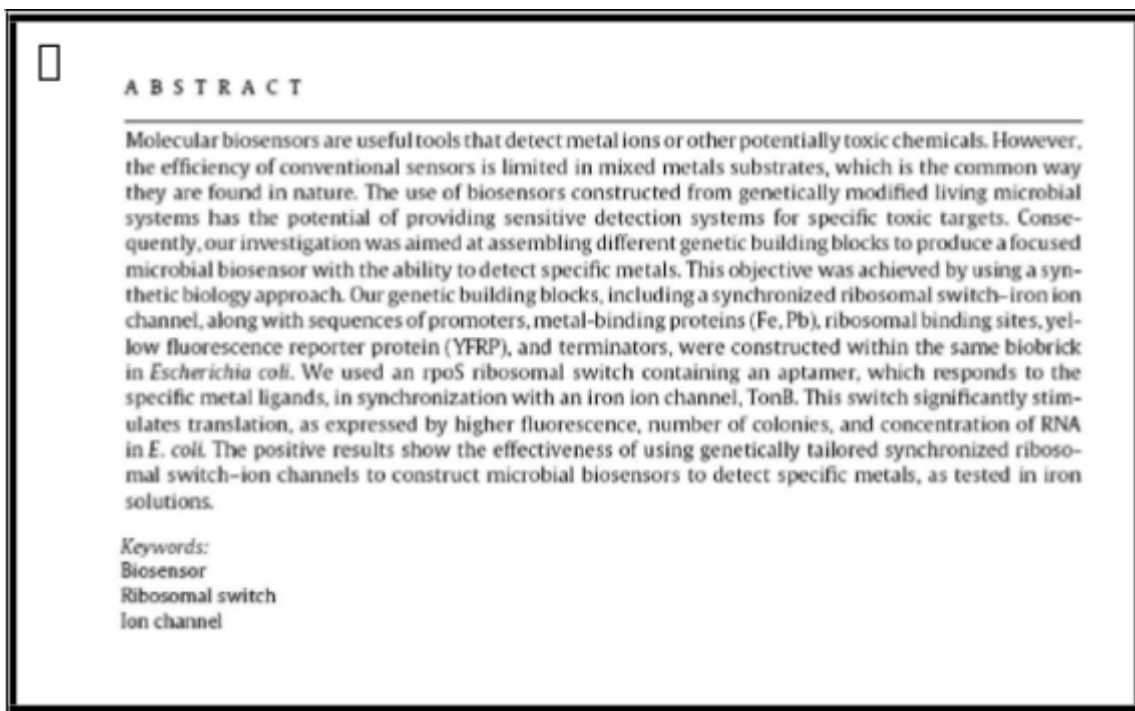
TITLE PAGE INFORMATION (see the example above)

- ❖ The title should be concise and informative as it will be used in information- retrieval systems. Avoid abbreviations and formulae where possible.
- ❖ Author names and affiliations: where the family name may be ambiguous (e.g., a double name), please indicate this clearly. Your official affiliation address is "School of Biotechnology, International University – Vietnam National University in HCMC". Indicate all affiliations with a lower-case superscript letter immediately

- ❖ after the author's name and in front of the appropriate address if your adviser/co-worker is from another institution. Provide the e-mail address of the corresponding author, i.e. yours in most cases.

ABSTRACT

- ❖ **250 - 300 words.** And should be as a single paragraph.
- ❖ **Keywords:** immediately after the abstract. Provide a maximum of 6 keywords, using American spelling and avoiding general and plural terms and multiple concepts (avoid, for example, 'and', 'of'). Be sparing with abbreviations: only abbreviations firmly established in the field may be eligible. These keywords will be used for indexing purposes.



TABLES

- ❖ Number tables consecutively in accordance with their appearance in the text.
- ❖ **Table legends should be placed above the table body.**
- ❖ Place footnotes to tables below the table body and indicate them with superscript lowercase letters. Avoid vertical rules.
- ❖ Be sparing in the use of tables and ensure that the data presented in tables do not duplicate results described elsewhere in the article.

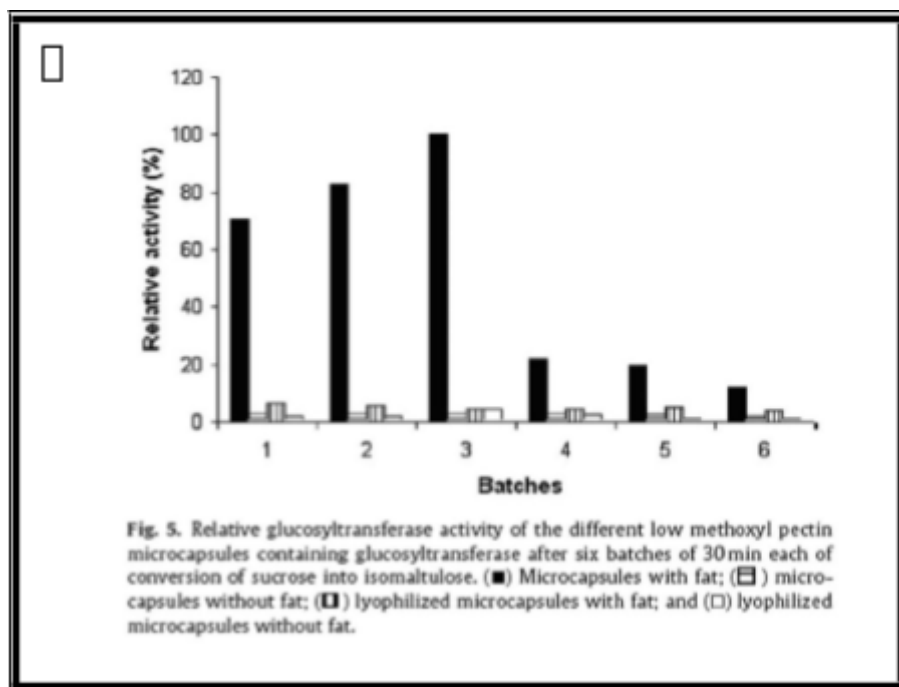
Examples:

Assay	Variables			Conversion of sucrose into isomaltulose (%)		
	pH	Enzyme (U/g of Celite)	Glutaraldehyde (%)	1 st batch	2 nd batch	3 rd batch
1	-1 (5.6)	-1 (32.6)	-1 (0.10)	7.38	7.38	9.03
2	+1 (7.4)	-1 (32.6)	-1 (0.10)	0.00	0.00	0.00
3	-1 (5.6)	+1 (87.0)	-1 (0.10)	21.92	21.92	23.63
4	+1 (7.4)	+1 (87.0)	-1 (0.10)	1.34	1.34	1.59
5	-1 (5.6)	-1 (32.6)	+1 (0.40)	1.51	0.00	1.59
6	+1 (7.4)	-1 (32.6)	+1 (0.40)	0.00	0.00	0.00
7	-1 (5.6)	+1 (87.0)	+1 (0.40)	12.75	8.73	10.64
8	+1 (7.4)	+1 (87.0)	+1 (0.40)	0.00	1.52	1.15
9	-1.68 (5.8)	0 (59.8)	0 (0.25)	19.81	18.09	20.32
10	+1.68 (8.6)	0 (59.8)	0 (0.25)	0.00	0.00	0.00
11	0 (6.5)	-1.68 (14.1)	0 (0.25)	0.00	0.00	0.00
12	0 (6.5)	+1.68 (105.5)	0 (0.25)	7.23	8.00	7.19
13	0 (6.5)	0 (59.8)	-1.68 (0.00)	16.94	14.12	11.54
14	0 (6.5)	0 (59.8)	+1.68 (0.50)	3.25	2.87	3.77
15	0 (6.5)	0 (59.8)	0 (0.25)	4.31	6.33	4.62
16	0 (6.5)	0 (59.8)	0 (0.25)	6.18	5.96	4.29

FIGURE CAPTION

Ensure that each illustration has a caption. Figure legend should be placed below the Figure. A caption should comprise a brief title and a description of the illustration. Keep text in the illustrations themselves to a minimum but explain all symbols and abbreviations used.

Example:



Note:

General points for Electronic artwork: (Table, Figure...)

- Use uniform lettering and sizing of your original artwork.
- Embed the used fonts if the application provides that option.
- Aim to use the following fonts in your illustrations: Arial, Courier, Times New Roman, Symbol, or use fonts that look similar.

CITATION IN TEXT

Please ensure that every reference cited in the text is also present in the reference list and vice versa. Any references cited in the abstract must be given in full. Unpublished results and personal communications are not recommended in the reference list, but may be mentioned in the text. If these references are included in the reference list they should follow the standard reference style as follows and should include a substitution of the publication date with either 'Unpublished results' or 'Personal communication'. Citation of a reference as 'in press' implies that the item has been accepted for publication.

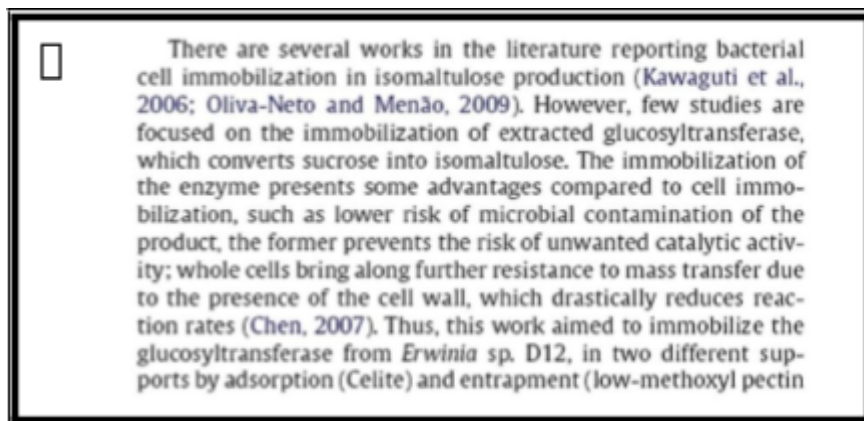
All citations in the text **should** refer to:

- *Single author*: the author's name (without initials, unless there is ambiguity) and the year of publication;
- *Two authors*: both authors' names and the year of publication;
- *Three or more authors*: first author's name followed by 'et al.' and the year of publication.

Citations may be made directly (or parenthetically). Groups of references should be listed first alphabetically, then chronologically.

Citation may be displayed in numeric format, too. Ex: There are several works in the literature reporting bacterial cell immobilization in isomaltulose production [1].

Use uniform format for your citation and Reference list.



WEB REFERENCE

As a minimum, the full URL should be given and the date when the reference was last accessed. Any further information, if known (DOI, author names, dates, reference to a source publication, etc.), should also be given. Web references can be listed separately (e.g., after the reference list) under a different heading if desired, or can be included in the reference list. Avoid using websites as reference unless absolutely necessary.

REFERENCE LIST

References should be arranged first alphabetically and then further sorted chronologically if necessary. More than one reference from the same author(s) in the same year must be identified by the letters 'a', 'b', 'c', etc., placed after the year of publication. Journal name must be written in full name. Student should use Citation software to make this section uniform (EndNote, Mendeley... are commonly used in scientific writing).

Examples:

Reference to a journal publication:

Van der Geer, J., Hanraads, J.A.J., Lupton, R.A., 2010. The art of writing a scientific article. *Journal of Science Communication* 163, 51–59.

Reference to a book:

Strunk Jr., W., White, E.B., 2000. *The Elements of Style*, fourth ed. Longman, New York.

Reference to a chapter in an edited book:

Mettam, G.R., Adams, L.B., 2009. How to prepare an electronic version of your article, in: Jones, B.S., Smith, R.Z. (Eds.), *Introduction to the Electronic Age*. E-Publishin.

References

Andrianantoandro, E., Basu, S., Karig, D.K., Weiss, R., 2006. Synthetic biology: new engineering rules for an emerging discipline. *Molecular Systems Biology* 2 (28), 1–14.

Breaker, R.R., 2010. RNA second messengers and riboswitches: relics from the RNA world. *Microbe American Society for Microbiology* 5 (1), 13–20.

Cuero, R., Ouellett, T., Yu, J., Mogongwa, N., 2003. Metal ion enhancement of fungal growth, gene expression, and aflatoxin synthesis in *Aspergillus flavus*: RT-PCR characterization. *Journal of Applied Microbiology* 94 (6), 953–961.

Cuero, R., Ouellett, T., 2005. Metal ions modulate gene expression, and accumulation of the mycotoxins aflatoxin and zearalenone. *Journal of Applied Microbiology* 98 (3), 598–605.

Failla, M.L., 1977. *Zinc Functions and Transport in Microorganisms*, 4th ed. Weinberg, New York.

Grundy, F.J., Henkin, T.M., 2006. From ribosome to riboswitch: control of gene expression in bacteria by RNA structural rearrangements. *Critical Reviews in Biochemistry and Molecular Biology* 41 (6), 329–338.

Hengge-Aronis, R., 2002. Signal transduction and regulation mechanisms involved in control of the sigma (s) RpoS subunit of RNA polymerase. *Microbiology and Molecular Biology Review* 66 (3), 373–395.

Hille, B., 2001. *Ion Channels of Excitable Membranes*, 3rd ed. Sinauer, Sunderland.

Ito, M., Xu, H., Gufanti, A.A., Wei, Y., Zvi, L., Clapham, D.E., Krulwich, T.A., 2004. The voltage-gated Na⁺ channel NavBP has a role in motility, chemotaxis, and pH homeostasis of an alkaliphilic *Bacillus*. *Proceedings of the National Academy of Sciences* 101 (29), 10566–10571.

Kauffman, S., 2000. *Investigations*. Oxford University Press, New York.

Lei, Y., Chen, W., Mulchandani, A., 2006. Microbial biosensors. *Analytica Chimica Acta* 568 (1), 200–210.

Mijakovic, I., 2010. Protein phosphorylation in bacteria. *Microbe ASM News* 5 (1), 21–25.

Nudler, E., Mironov, A.S., 2004. The riboswitch control of bacterial metabolism. *Trends in Biochemical Science* 29 (1), 11–17.

APPENDIX

All materials placed in the appendix must be directly relevant to the paper. The material must be cross-referenced to the development of the research in the text of the paper using an explanatory note or a parenthetical reference. Avoid the temptation to use the appendix to bulk up the paper.

LANGUAGE AND GRAMMAR

- Use simple but clear language
- Take time to check your work for misspelled words, typographical error, mislabelled figures, tables or photos.

- If you need help in grammar, seek the help of an editor before submitting your work to your adviser. Your adviser is not expected to correct errors in spelling, punctuation, grammar, and formatting.

ABBREVIATION

Define abbreviations that are not standard in this field in a footnote to be placed on the first page of the article. Such abbreviations that are unavoidable in the abstract must be defined at their first mention there, as well as in the footnote. Ensure consistency of abbreviations throughout the article.

ACKNOWLEDGING THE WORK OF OTHERS

Plagiarism

Plagiarism is copying another person's idea or written work and claiming it as your own. This is an academic offence and you are strictly prohibited from doing this. Make sure that all information, photos, figures and tables are properly acknowledged

Citations

You must always acknowledge your sources of factual information and diagrams you wish to use. This is known as a *citation*.

PART 4: THESIS DEFENCE

PRESENTATION

- Presentation should last up to 15 minutes with another 15 minutes for questions and answers
- Slides should be prepared using Microsoft PowerPoint and presented from a disk.
- Rehearse your presentation and anticipate questions that may be asked by the Evaluation Committee.
- If you are not sure about the pronunciation of certain terminologies, be sure to ask a knowledgeable person before your defence.
- Try not to read from your slides and maintain eye contact with your audience
- Use pointers or laser devices properly
- Ask your supervisor for advice on the content and structure of your presentation.
- Even a successful defense is generally followed by certain minor adjustments in your document, and some final paperwork amendments. You should take notes during the Q&A session, and contact the Secretary of the Evaluation Committee for a detailed request for thesis improvement.

CONTENT OF PRESENTATION

- The presentation should be a brief introduction of your topic, purpose of your study; description of the methods used and the results.
- It is advisable that your presentation has enough important details in order to avoid misunderstanding or excessive questions. Also, keep it short as time is limited.
- Make sure your answers are relevant to the questions of the Evaluation Committee.

APPENDIX 1: FORMAT OF THESIS COVER PAGE

VIETNAM NATIONAL UNIVERSITY – HOCHIMINH CITY
INTERNATIONAL UNIVERSITY



(5 lines from logo)

TITLE OF THESIS

(3 lines)

A thesis submitted to
The School of Biotechnology, International University
In partial fulfillment of the requirements for the degree of
B.S. in

(6 lines)

Student name: Full name of student – ID No.

Supervisor: Title and full name of supervisor(s)

(7 lines)

APPENDIX 2: RELEVANT FORMS

(proposal development, proposal defense, midway progress report, evaluation, etc.)

Content	Page
Form No 1: Thesis registration	19
Form No 2: Thesis progress report	20
Form No 3: Academic Adviser	22
Form No 4: Thesis Reviewer	23
Form No 5: For Examiner Of The Scientific Committee	24
Form No 6: Thesis Evaluation Memo	25
Form No 7: Report on thesis revision	27



TRƯỜNG ĐẠI HỌC QUỐC
TẾ KHOA CÔNG NGHỆ
SINH HỌC

INTERNATIONAL UNIVERSITY
SCHOOL OF BIOTECHNOLOGY

Form BT01

ĐĂNG KÝ ĐỀ TÀI TỐT NGHIỆP THESIS REGISTRATION

1. Họ và tên SV (Student's name) MSSV (ID)
2. Bộ môn thực hiện (Department)
3. Tên đề tài (Thesis title)
.....
.....
4. Mục tiêu (Objectives)
.....
.....
.....
5. Nội dung nghiên cứu (Research content)
.....
.....
.....
6. Địa điểm nghiên cứu (Research location)
.....
7. Thời gian thực hiện (Duration) từ (from): đến (to):
8. Người hướng dẫn (Supervisor):
Họ tên (Full name)
Địa chỉ (Address).....
.....
Email:

Phê duyệt của GVHD
(Supervisor)

TPHCM, ngày tháng năm
Sinh viên
(Ký và ghi rõ họ tên)

Phê duyệt của Bộ môn
(Department)

INTERNATIONAL UNIVERSITY
School of Biotechnology



THESIS PROGRESS REPORT

1. Student name: Student's ID.....

2. Supervisor

3. Thesis title

SECTION A: to be completed by student

Thesis processing management

Content	Status		Tentative completion time
	Complete	On going	
1.	☐	☐	
2.	☐	☐	
3.	☐	☐	
n.	☐	☐	

Presence of obstacles to thesis completion, if any,

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Important note: Date to submit the completed thesis:

Date:

Signature of student

SECTION B: to be completed by the principal Supervisor

Has the student:	Yes	No
(i) Shown relevant knowledge and understanding toward specific project field?	☐	☐
(ii) Shown initiative consistent with the requirements of the research program?	☐	☐
(iii) Made satisfactory progress in the research program?	☐	☐
(iv) Shown the ability to complete the research program by the due date?	☐	☐

If no, please recommend extension for completion or cut some parts of the proposal

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Evaluation Form

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Name of Student: ID:

Criteria	Maximum marks	Your mark
Independence in work	10	
Creativity	10	
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Overall quality of thesis *	40	
Total	100	

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Evaluation Form

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Writing skill and format (including compliance do thesis guidelines)	15	
Total	100	

Comments and recommendations for improvement/ correction (blank section is not acceptable)

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Name of Reviewer (Signature and Date)

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Evaluation Form

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Introduction (<i>research problem well stated, clear objectives</i>)	10	
Good understanding of the research field	10	
Methodology (<i>sound, appropriate or creative</i>)	20	
Quality of results (<i>evaluated against the research objectives</i>)	20	
Presentation skills (<i>quality of slides, speaking skills, timing</i>)	20	
Quality of answers (<i>relevant to questions, satisfied by the committee members</i>)	20	
Total	100	

Additional comments/suggestions for improvement:

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Name of Examiner

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THESIS EVALUATION MEMO **BIÊN BẢN ĐÁNH GIÁ LUẬN VĂN TỐT NGHIỆP**

1. Thesis topic (Tên đề tài):
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2. Student's name (Người thực hiện):
3. Defence date (Ngày bảo vệ):
4. The committee (Tại hội đồng): – School of Biotechnology (Khoa Công nghệ Sinh học)
5. Evaluation of Scientific committee (Điểm đánh giá của Hội đồng)

No. STT	Member of the committee Thành viên	Score Điểm
1		
2		
3		
4		
5		
6	Thesis advisor (Người hướng dẫn)	
7	Thesis reviewer (Người phản biện)	
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Recommendation for thesis revision: (Các ý kiến kết luận của Hội đồng để chỉnh sửa luận văn):

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[illegible]

The committee hereby assigns the advisor to supervise and make sure all recommendations are well addressed by the student. The revised thesis must be submitted to the School's secretary by . (Hội đồng giao cho giảng viên hướng dẫn có trách nhiệm hướng dẫn và kiểm tra các chỉnh sửa trên trước khi sinh viên nộp bản luận văn đã chỉnh sửa cho thư ký Khoa trước ngày)

Secretary of the committee

Thư ký hội đồng

Chair of the committee

Chủ tịch hội đồng

REPORT ON THESIS REVISION



Name of Student: ID:

Thesis title:

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No.	Recommendation	Revision	Page
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Date of submission:

Student's signature:

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Approved by Chairman (or the Authorized person) :

The undersigned certifies that I have approved the revision made by student.

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APPENDIX 3: SAMPLE PAPERS

(two sample papers collected from Journal of Biotechnology, Elsevier)



Constructed molecular sensor to enhance metal detection by bacterial ribosomal switch–ion channel protein interaction

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Molecular biosensors are useful tools that detect metal ions or other potentially toxic chemicals. However, the efficiency of conventional sensors is limited in mixed metals substrates, which is the common way they are found in nature. The use of biosensors constructed from genetically modified living microbial systems has the potential of providing sensitive detection systems for specific toxic targets. Consequently, our investigation was aimed at assembling different genetic building blocks to produce a focused microbial biosensor with the ability to detect specific metals. This objective was achieved by using a synthetic biology approach. Our genetic building blocks, including a synchronized ribosomal switch–iron ion channel, along with sequences of promoters, metal-binding proteins (Fe, Pb), ribosomal binding sites, yellow fluorescence reporter protein (YFRP), and terminators, were constructed within the same biobrick in *Escherichia coli*. We used an rpoS ribosomal switch containing an aptamer, which responds to the specific metal ligands, in synchronization with an iron ion channel, TonB. This switch significantly stimulates translation, as expressed by higher fluorescence, number of colonies, and concentration of RNA in *E. coli*. The positive results show the effectiveness of using genetically tailored synchronized ribosomal switch–ion channels to construct microbial biosensors to detect specific metals, as tested in iron solutions.

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1. Introduction

Biosensors are useful cellular tools that allow the detection of a wide range of elements, in many cases overcoming the limitations of conventional methods. Although in nature metals are found in mixtures, multiple compounds, and ionized species, many conventional sensors fail to detect specific metals (Cuero et al., 2003; Cuero and Ouellet, 2005). Metal ions have been shown to be essential in the synthesis of nucleic acids (Failla, 1977) where the ligand binds with the nucleic acids (Cuero et al., 2003; Cuero and Ouellet, 2005). Through the introduction and coordination of multiple genes, biological systems have been designed to specifically identify target molecules, and these systems are termed biosensors (Prow et al., 2004). Such biosensors can be used to identify specific metals, which in nature are found in mixtures (Cuero et al., 2003; Cuero and Ouellet, 2005).

Heavy metals are an increasing pollutant of the environment and there is a growing need to develop sensitive and selective methods for their detection. Although metals enter the cell through

different mechanisms, ion channels are common paths for metals to enter the cell. Voltage-gated ion channels are proteins embedded in the cellular membrane that allow the passage of ions into a cell through an electrochemical gradient and are target selective based on the tertiary structure of the protein (Hille, 2001). Different types of ion channels are found in bacterial cells; however, the voltage-gated type of ion channel seems to play important survival roles such as chemotaxis and homeostasis in prokaryotes (Ito et al., 2004). The voltage-gated Ton Box (TonB) ion channel allows the passage of bivalent cations, such as Fe (II), which was assembled for selective passage through the cellular membrane into the cell, oxidizing the ions as they enter. Some bacteria, such as *E. coli*, can exhibit natural protein receptors at the periplasm level, such as TonB, which could function as ion channels (Quintero et al., 2007). The Na⁺ ion has been used in a voltage-gated channel in bacteria, showing a reciprocal modulatory effect on chemosensory transduction (Ito et al., 2004). The integration of ion channels into an artificial polymer membrane in combination with an optical sensor has been considered for drug screening and biochemical analysis (Steiner et al., 2003). An allosteric ribozyme has been engineered to respond to specific divalent metal ions (Zivarts et al., 2005).

Ribosomal switches are structured RNA domains found in many mRNAs of bacterial species that detect small molecules and control the expression of associated genes (Breaker, 2010). This

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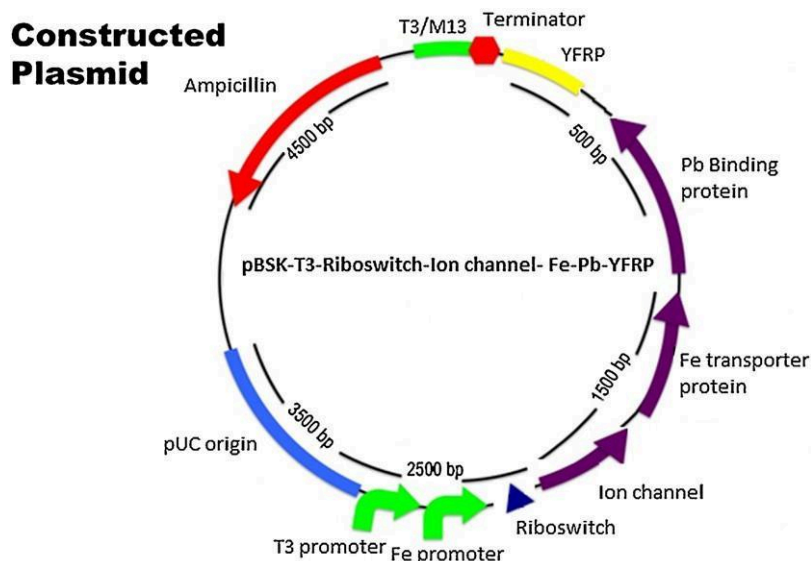


Fig. 1. Graphic illustration of constructed plasmid showing the direction, placement and size of genes within the pBSKII vector beginning with the T3 promoter, iron promoter, riboswitch rpoS, ion channel, iron-binding protein, lead-binding protein, yellow fluorescent reporter protein (YFRP), and the terminator sequence. Ampicillin resistance gene is included as a genetic marker and was used for screening colonies that had successfully been transformed with the plasmid.

function suggests that domain-ligand binding may form between RNA and metal ions. Riboswitches are a two-part system consisting of an aptamer, which changes conformation when bound to its metabolite, and an expression platform containing the regulated genes. The regulation of gene expression on a translational level can be mitigated by mRNA riboswitch folding that responds to small target molecules either activating or deactivating the translation of downstream genes (Nudler and Mironov, 2004). Selective riboswitches in response to environmental conditions in the presence of metal contamination can be used to control gene expression. The untranslated 5' aptamer of the rpoS riboswitch binds to the stationary-phase sigma factor S1, which has been identified as a survival response to extreme environmental conditions, such as oxidative stress (Vitreschak et al., 2004). Based on such examples, the use of synchronized riboswitch-ion channels assembled with different gene building blocks has been used advantageously to detect specific metals (Quintero et al., 2007).

Synthetic biology is a developing scientific research approach that coordinates engineering and classical molecular genetics techniques to develop biological systems with specific functions (Andrianantoandro et al., 2006). This tool can be applied to the design, understanding, and operation of genetic systems through the assemblage of genes derived from a wide range of different organisms. Synthetic biology provides the ability to design biological systems with specific functions that do not exist in nature. It provides tools to engineer mechanisms for controlling genetic expression at the regulatory level, including detection of specific metals. The use of synthetic biology to develop biosensors for detecting Fe^{2+} was reported by Quintero et al. (2007).

A microbial biosensor is an analytical device that is able to accurately and sensitively detect specific targets in different fields, such as environmental monitoring, medicine, food processing, and food safety. Many of these microbial biosensors have been engineered based on the assemblage of synthetic genetic parts (Lei et al., 2006). However, despite the effective assemblage of the genetic parts or construction of the microbial biosensor, the metabolic activity is not good enough to express sensitive detection (Quintero et al., 2007).

In this study we apply concepts of synthetic biology in combination with conventional scientific methods to assemble different

genetic components to construct a bacterial molecular biosensor for enhanced detection of metals. This work combines sequences of different promoters, metal-binding proteins (Fe^{2+} , Pb^{2+} , Na^+ , and Cl^-), ribosomal binding sites, fluorescent reporter protein, terminators, and a synthetic riboswitch, in synchronization with a protein (TonB) playing the role of an iron-ion channel.

The objectives of this study were to develop a biosensor engineered and assembled for potential sensitive detection of bivalent cations such as iron and lead, and to use a ribosomal switch to regulate the specific detection of these metals. The detection of these cation-binding proteins (Fe and Pb) would signal the properties of *E. coli* colonies that have been modified to respond to them and would increase the fluorescence of the protein reporter.

2. Materials and methods

2.1. Bacterial culture and competent cells

Bacterial strains of *E. coli* DH10 were obtained from Invitrogen (USA), grown in Luria-Bertani (LB) medium, and incubated at 37 °C. These bacterial strains were used as competent cells in which different genetic parts were assembled.

2.2. Genetic parts

The DNA biosensor device was assembled in *E. coli* competent cells to construct the molecular bacterial metal sensor. The rpoS ribosomal switch and TonB transporter ion channel protein sequences were obtained from Gene Bank: D26134.1 and ACB93044.1, respectively. The riboswitch rpoS gene was synthesized with the addition of Hind III & Kpn I sites at 5' and Hind III & Xho I sites at 3'. The synthetic gene was sub-cloned into EcoRV site of pBSK II vector. Other synthesized gene parts were also assembled into the device, including Fe promoters, metal-binding proteins (Fe, Pb), yellow fluorescent reporter protein (YFRP), and a ribosomal binding site (RBS) J61100 taken from the iGEM registry parts (MIT, MA). The sequences of these gene parts were also obtained from the Gene Bank in the Pubmed database and synthesized by CloneTEX company (Austin, TX).

Different genetic parts and/or sequences were tested and purified through DNA extraction and electrophoresis. DNA extraction

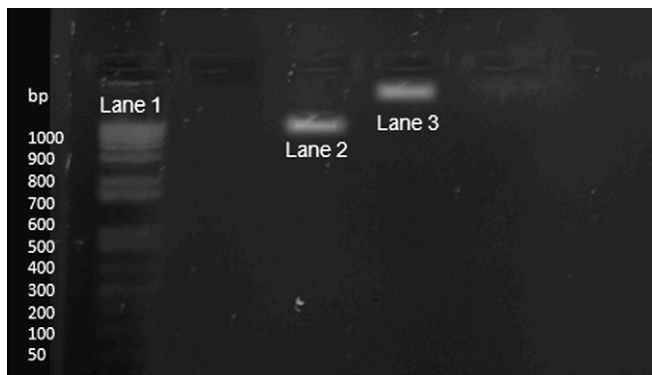


Fig. 2. Gel of DNA of the two biosensor devices grown without iron, after electrophoresis. The photo shows, from left to right, lane 1, DNA marker (1000 bp); lane 2, DNA of biosensor device without ribosomal switch showing lower molecular weight than the biosensor device with ribosomal switch; lane 3, DNA of biosensor device with ribosomal switch showing higher molecular weight than biosensor without ribosomal switch.

was performed from single bacterial colonies using WIZARD Plus Minipreps DNA Extraction and Purification System (Promega), followed by electrophoresis.

2.3. Enzyme digestion, assemblage of biosensor, and ligation

Digestion reactions were carried out at 37 °C using New England Biolab restriction endonuclease enzymes and buffers, and each single gene was digested with compatible restriction sites from the multiple cloning region of the pBSKII plasmid. Using 4 µl of 10 buffer, 4 µl of the template DNA, 0.4 µl BSA, 2 µl of each restriction enzyme, and 27.6 µl ddH₂O, a final volume of 40 µl was incubated on a heat block at 37 °C for 2 h to ensure complete digestion. Alkaline phosphatase treatment was added one hour into incubation. Samples were then run in an electrophoresis gel to separate the DNA fragments using a molecular ladder, allowing identification of the gene of interest. Gel bands were isolated and purified using the NEB gel purification kit.

The assemblage of genes was carried out according to Biobrick protocol into a pBSKII high-copy plasmid, and selected genes were ligated into the multiple cloning region of the pBSKII plasmid. After the different parts were tested and/or purified, they were assembled in a proper sequential order in competent *E. coli* cells, thus constructing the DNA biosensor device. The sequential

order was as follows: T3 promoter + Fe-promoter + Ribosomal switch rpoS + Ion channel TonB + Fe-binding protein + Pb-binding protein + yellow fluorescence protein reporter + terminator (Fig. 1).

Ligations were performed following the Quick Ligation Protocol of New England BioLabs using T4 ligase. Fifty ng of backbone vector were combined with a 3-fold molar excess of gene insert. Volume was adjusted to 10 µl using ddH₂O, after which 10 µl of

Quick Ligation Buffer was added, along with 1 µl of T4 DNA Ligase; the combination was mixed thoroughly. Each reaction was centrifuged briefly and incubated at room temperature (25 °C) for 5 min before transformation of competent cells or storage at 20 °C.

Two types of DNA biosensor devices were constructed, either with or without the ribosomal switch sequence. Each type of device was compared to the control, which consisted of non-transformed competent *E. coli* DH10 cells.

2.4. RNA extraction

Bacterial cell cultures were harvested at 24 h and subjected to RNA extraction, following QIAGEN protocol (CA, USA). Bacterial cells were harvested by centrifuging 2 mL of cell culture at 5000 *g* for 5 min at 4 °C and decanting the supernatant. Then 350 µl of Buffer RLT was added and resuspended by vortexing for 10 s. After transferring 700 µl of lysate to an RNeasy spin column, the material was placed into a 2 mL tube, the lid was closed gently, and the mixture was centrifuged at 8000 *g* for 15 s. The flow-through was poured off, and 700 µl of Buffer RW1 was added to the spin column; again, the lid was closed gently and the mixture was centrifuged at 8000 *g* for 15 s. The flow-through was poured off again and 500 µl of Buffer RPE was added to the spin column, the lid closed gently, and centrifuged at 8000 *g* for 15 s. The flow-through was poured off a third time and the above washing step was repeated. The spin column was placed into a new 2 mL tube, 40 µl of RNase free water was added to the spin column membrane, and the mixture was centrifuged at 13,000 *g* for 1 min. The RNA concentration of the device or transformed cells of the biosensor were quantified using triplicate 1-mL samples and measured in a nano-spectrophotometer (nanoDrops-UV/vis spectrophotometer-MET-Limited), for each iteration.

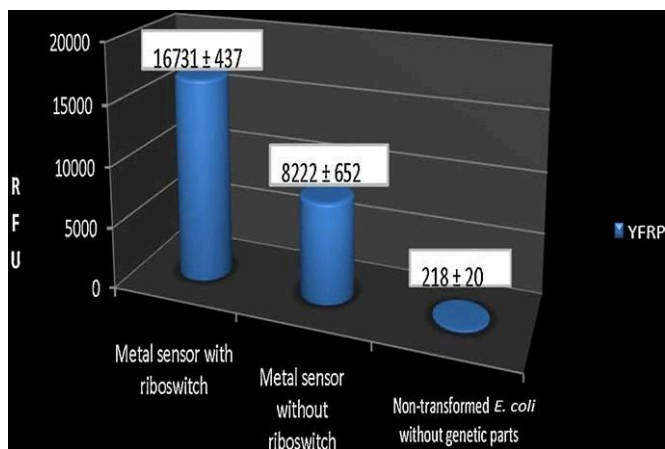


Fig. 3. The detection ability of developed biosensor, grown without iron, showing measured relative fluorescent units (RFU) among the biosensor for metals with riboswitch translational regulation of YFRP, biosensor without ribosomal control, and control (non-transformed) *E. coli* cultures.

Fig. 4. The detection ability of developed biosensor, grown without iron, showing colony-forming units and RNA concentrations of the biosensor for metals with riboswitch translational regulation of the constructed device, biosensor without ribosomal control, and control (non-transformed) *E. coli* cultures.

CFU X 10⁶

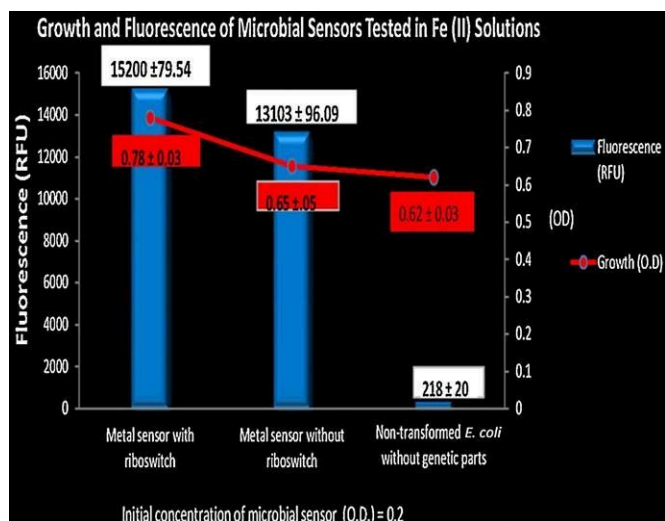


Fig. 5. The detection ability of developed biosensor, grown in iron (5 µg/ml), showing growth and fluorescence (RFU) of the biosensor for metals with riboswitch translational regulation of the constructed device, biosensor without ribosomal control, and control (non-transformed) *E. coli* cultures.

2.5. Fluorescence measurement

Fluorescence of the bacterial sensor was determined after 24 h of growth using a 20/20 luminometer (Turner Biosystems, USA). Triplicate 100-µl samples of bacterial sensor were used for determining fluorescence for each iteration, and the results were expressed in relative units of fluorescence (RFU).

2.6. Test of bacterial sensor in metal solution

Detection efficacy of the bacterial sensor was measured in iron or lead sulfate solutions at 5 µg/ml concentration. These tests were done using bacterial sensors at concentrations of 0.2 optical density (O.D.), measured by standard spectrophotometry method, UV/vis spectrophotometer (BECKMAN, USA).

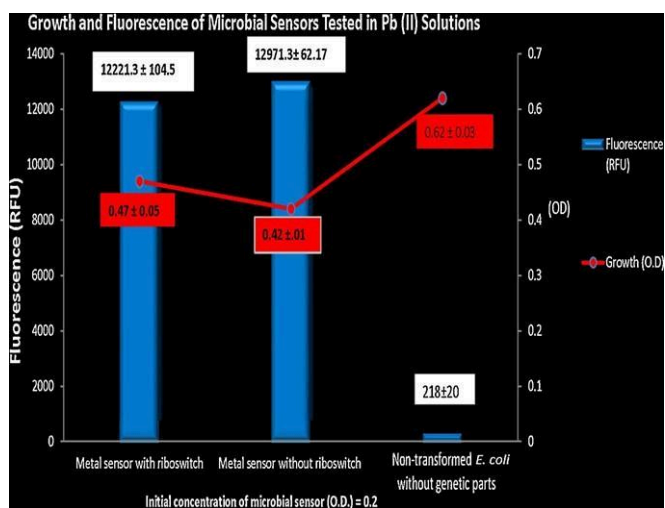


Fig. 6. The detection ability of developed biosensor, grown in lead (5 µg/ml), showing growth and fluorescence (RFU) of the biosensor for metals with ribosomal switch translational regulation of the constructed device, biosensor without ribosomal control, and control (non-transformed) *E. coli* cultures.

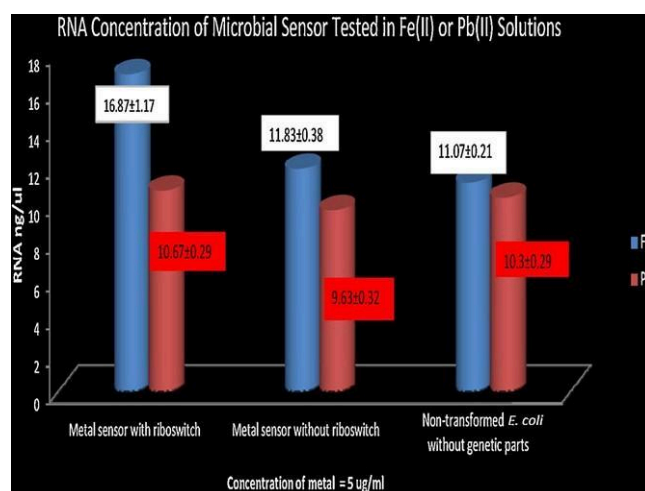


Fig. 7. The detection ability of developed biosensor, grown in iron or lead (5 µg/ml), showing RNA concentrations of the biosensor for metals with riboswitch translational regulation of the constructed device, biosensor without ribosomal control, and control (non-transformed) *E. coli* cultures.

3. Results

Effective assemblage of the microbial biosensor was confirmed by the tested ribosomal switch-ion channel synchronization in relation to the iron and lead protein interaction and the level of fluorescence of the yellow reporter protein. Fig. 2 shows the single DNA band of the ligation of parts of the DNA metal biosensor device with and without a ribosomal switch after electrophoresis, corroborating the effective assemblage of the parts of the microbial biosensor device (gel for the Pb ion channel is not shown). This result shows the efficacy of the constructed bacterial biosensor with ion channel containing a ribosomal switch compared to the bacterial sensor with ion channel that did not contain a ribosomal switch. The effective assemblage of the parts of the microbial biosensor device is demonstrated in cases of both Fe and Pb ions; the presence of the target ion, specifically Fe, was efficiently detected by the DNA metal biosensor containing the ribosomal switch through analysis of measured parameters. This may indicate the effective interaction between the ribosomal switch and ion channel.

This detection was reflected by the increased growth (measured in CFU, O.D.) of the bacterial sensor (Figs. 4–6 and 8). Growth was higher for the bacterial sensor with the ribosomal switch-ion channel genes that were grown in an iron solution. These agar plates were fully covered by bacterial colonies (Figs. 5 and 8c), whereas this was not the case for the bacterial sensors without the ribosomal switch-ion channel genes (Fig. 8d), grown either with or without the lead solution (Fig. 6). Concentrations of RNA were also higher in the bacterial sensor containing the ribosomal switch-ion channel genes compared to the bacterial sensor without the ribosomal switch, regardless of whether they were grown in the iron solution, the lead solution, or with neither. Additionally, the RNA concentration was higher in the bacterial sensor grown in iron or lead solution compared to the bacterial sensor grown without iron or lead solution (Figs. 4 and 7). Perhaps the higher RNA concentration in the bacterial sensor with the ribosomal switch confirms that the ribosomal switch containing an aptamer, which responds to the iron ions acting as a ligand, synchronizes with the iron ion channel, TonB (Cuero et al., 2003; Cuero and Ouellet, 2005; Quintero et al., 2007) to enhance the specificity of the molecular biosensor. The slight decrease of RNA concentration in the bacterial sensor could be due to the effect of the iron solution in the bacterial cell. The iron may have caused an ionic cellular shock during the first short period of bacterial cell growth. Perhaps RNA concentration would increase

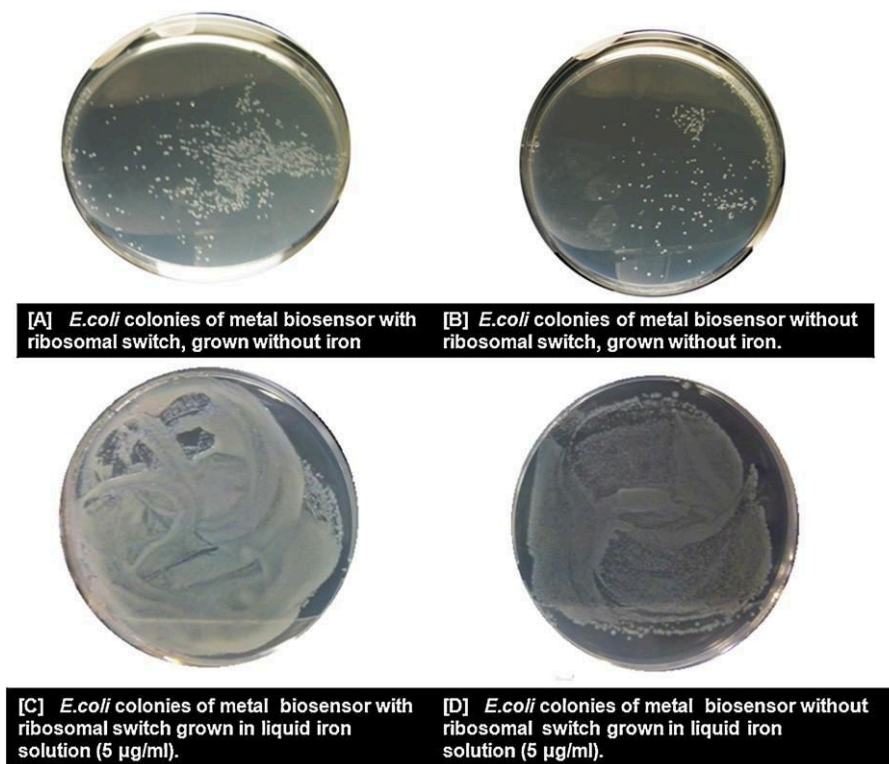


Fig. 8. *E. coli* colonies of metal biosensor with [A] or without [B] ribosomal switch, grown without iron. *E. coli* colonies of metal biosensor with [C] or without [D] ribosomal switch, grown in iron (5 µg/ml). Bacterial sensor colonies cover the surface of the agar plate, but bacterial sensor with ribosomal switch-ion channel grown in iron showed noticeably more growth.

again, after a longer period of bacterial growth, when the cell populations have adapted completely to the metal milieu (Cuero et al., 2003).

The bacterial sensor with ribosomal switch-ion channel, grown either with or without iron solution, also exhibited brighter fluorescence compared to the bacterial sensor without ribosomal switch, grown with or without lead solution (Figs. 3, 5, 6, 8a and c). This result corroborates the effect of the ribosomal switch in enhancing the detection of metal in the bacterial DNA metal sensor.

Fig. 4 exhibits a clear, direct correlation between concentration of RNA and bacterial population (CFU) of the device and the type of biosensor. The DNA metal biosensor with ribosomal switch showed higher CFU, corresponding to a higher RNA concentration, while the DNA biosensor without ribosomal switch showed both lower CFU and RNA concentration. These results also correlate to the fluorescence of the YFRP, expressed in relative fluorescence units (RFU); the DNA biosensor device containing the ribosomal switch showed higher RFU (16,731), followed by the DNA sensor without ribosomal switch (8227 RFU), and then the control, which showed the lowest fluorescence (218 RFU) (Fig. 3).

4. Discussion

The assemblage of the standardized biobrick parts with the synthesized sequences, such as metal-binding proteins, TonB ion channel, and rpoS ribosomal switch, was achieved effectively as reflected in their sensitivity to the metal ions. This effective assemblage of the biosensor parts is due to the appropriate design of the position and/or sequential organization of the parts within the device or biobrick (Quintero et al., 2007). The fact that the ligation showed a higher position of the corresponding DNA band indicates that all parts were assembled. Fig. 2 shows the electrophoresis gel corresponding to the metal biosensor device containing all the parts along with rpoS, thus demonstrating that all the parts of the device

were ligated. (Other biosensor devices were not subjected to gel electrophoresis).

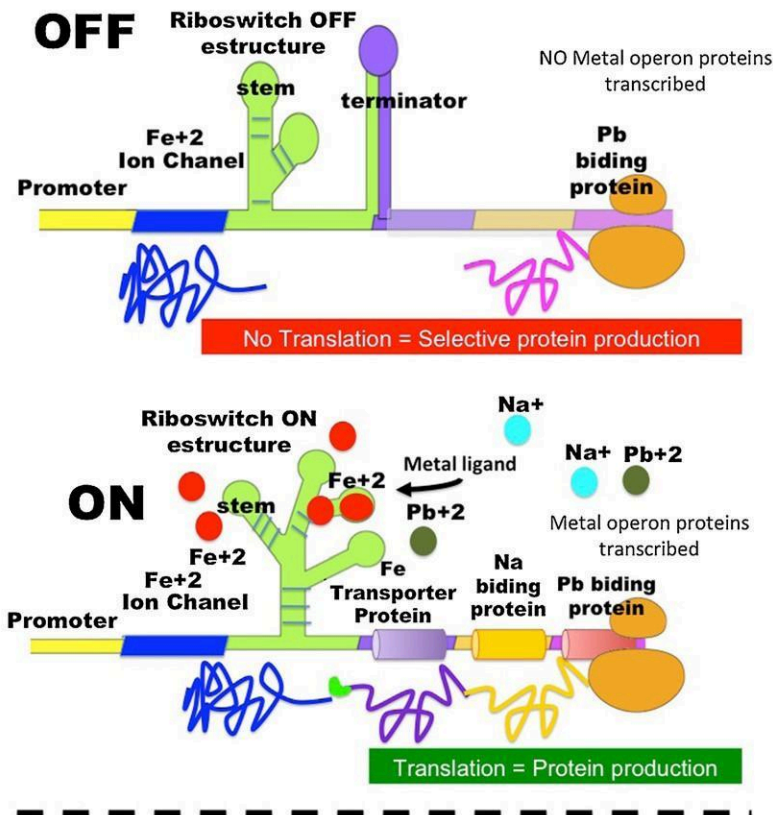
Genes are made of hundreds of atoms (Kauffman, 2000), therefore, it is rational to think that the interactions of the parts could be influenced by electrochemical principles of the atoms (Weast et al., 1989; Kauffman, 2000) at the binding sites.

Fig. 9. Graphic illustration of the ribosomal switch-ion channel synchronization. When bound to the S1 regulatory protein, the rpoS riboswitch undergoes a change in tertiary structure, allowing the ribosome to bind to the RBS and initiate translation of the downstream TonB ion channel. The translated voltage-gated ion channel becomes embedded in the cellular membrane, allowing bivalent cations to pass into the cell, becoming oxidized as they enter, aiding in the synthesis of ATP.

Fe²⁺

Ribosomalswitch / Ion channel synchronization favoring Iron (Fe²⁺) regulation as contrary to Pb(2+) regulation in *E. coli* cell.

- A.** Riboswitch ON: Favored Iron(2+) regulation. Pb regulation was not favored. Riboswitch OFF: Lead Pb(2+) regulation favored.



- B.** Riboswitch ON: Favored : Lead Pb(2+) regulation.) Iron(2+) regulation was not favored.

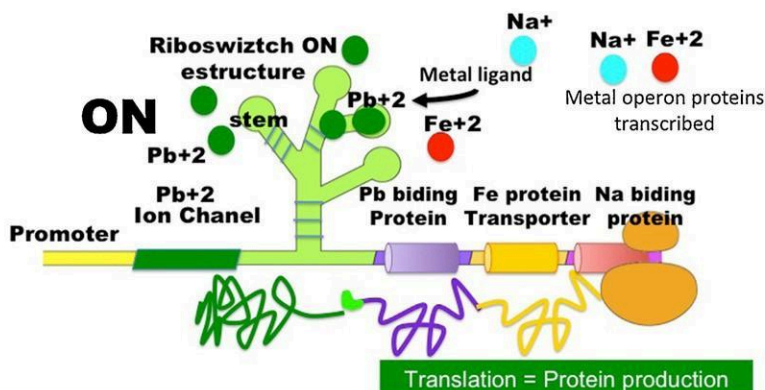


Fig. 10. Graphic illustration detailing the mechanics of the ribosomal switch-ion channel synchronization. (A) The change in rpoS tertiary structure when bound to ligand metabolite favoring iron (Fe²⁺) regulation. (B) Ribosomal switch favoring lead (Pb²⁺) regulation.

The correlation of selected different parameters, CFU, RNA, and intensity of the fluorescence (RFU) of the YFRP, measured the efficiency of the assembled biosensors in relation to the presence of the ribosomal switch. Fig. 4 demonstrates the correlation between the concentration of bacterial cells with the biosensor and the concentration of bacterial RNA; this shows a higher CFU (3×10^8 cells/ml) corresponding to the higher RNA concentration (105 ng/ μ l) in tested biosensors containing the ribosomal switch, and the lowest

CFU (1×10^7 cells/ml) corresponding to the lowest RNA concentration (22 ng/ μ l) in tested biosensors without the ribosomal switch. These results are confirmed by the brighter yellow fluorescence of the colonies seen in the biosensor containing the ribosomal switch, compared to the less bright fluorescence of the bacterial device without the ribosomal switch (Fig. 8) and no fluorescence of the colonies from the control without the assembled parts. Likewise, these fluorescence data correspond to fluorescence units (RFU) of

the YFRP, when determined photo-electrically in the 20/20 luminometer. Thus, higher fluorescence (16,731 RFU) was exhibited by tested metal biosensors with the ribosomal switch, compared to metal biosensors without the ribosomal switch (8222 RFU) and to control or bacterial cells without assembled parts (8222 and 218 RFU, respectively).

Metal ions have shown a direct effect on enhancing gene expression, and RNA concentration in microbial cells (Cuero et al., 2003; Cuero and Ouellett, 2005). However, the interaction of these metal proteins and their mode of regulation in relation to the detection of different types of metals were not previously elucidated. This synthetic biology approach has demonstrated the specific interactive effects of the ribosomal switch and ion channel proteins on regulating metal-binding protein expression by the biosensor (Figs. 3–8). These effects provide selective detection of the metal ions used in our system.

Results of this investigation clearly show the effect of the ribosomal switch in enhancing fluorescence of the biosensor reporter protein. It is likely that the ribosomal switch is mediating ion channel protein activity and/or regulating expression of the metal-binding proteins at the translation level. The ribosomal switch rpoS, which responds to oxidative stress such as the presence of metal ions, changes its tertiary structure when bound to the S1 sigma factor due to the methylation of its aptamer, and increases its affinity for ribosomes to bind with the ribosomal binding site immediately downstream (Hengge-Aronis, 2002). This translation regulation of the production and function of specific proteins seems to be mediated through synchronization between the ion channel TonB and ribosomal switch rpoS (Figs. 9 and 10).

Our results show higher fluorescence of the YFRP, as well as a higher number of fluorescent biosensor colonies containing the ribosomal switch, compared to biosensors without the ribosomal switch (Figs. 3–6 and 8). This suggests that the ribosomal switch in synchronization with the ion channel regulates expression and/or signaling of the metal-binding proteins.

Additionally, these signals may reflect an oxidative stress, which may have turned on the translation domain of the ribosome. However, this could depend on the availability and/or types of metal-binding proteins (Figs. 9 and 10). Bacterial RNA binding proteins with sensing properties in the regulatory system have been recognized. Ribosomal protein operons are repressed by individual ribosomal proteins when the amount of free proteins exceeds the amount of free ribosomal proteins. Consequently, RNA will exhibit signals that more production of ribosomal proteins is not needed (Grundy and Henkin, 2006). This may also explain how the synchronization of this ribosomal switch with an ion channel such as TonB had an effect on the mechanism of the metal-binding protein expression in our experiments.

Some of the regulatory elements of the RNA structure that affect the initiation region of translation or the target gene under normal growth conditions may unfold to permit a downstream translational sensing response (Grundy and Henkin, 2006). This seems to be the case in the present investigation with metal-binding proteins, in which the bacterial sensor with the ribosomal switch and ion channel protein showed more colony populations and higher protein fluorescence (Figs. 3–6 and 8).

This observation also confirms the successful assemblage of the genetic parts of the biosensor, as well as the expected increased metabolism of the bacterial cells containing the ribosomal switch, in which there was a corresponding higher production of RNA and translated proteins. This result was corroborated by both higher numbers of bacterial biosensor colonies and higher fluorescence of the YFRP (Figs. 3–6 and 8). Certain amino acids, histidine and

aspartate, are phosphorylated to form a two-component system. The first system is an input-sensing histidine-kinase that responds to specific stimulus. Then the kinase transfers the acquired phosphate to a residue aspartate, thus creating a second component. Sometimes the two components become integrated into a single system and are able to detect more complex signaling. In the same way, the biosensor enables bacterial cells to detect various environmental inputs, such as pH, nutrients, salinity, and temperature (Mijakovic, 2010).

We are carrying out further experiments to test the sensitivity of the different biosensors constructed, using varying concentrations of different types of metal ions. Also, new biosensors containing different ion channels and ribosomal switches are being assembled. One of the objectives of this new investigation is to synchronize ion channels with ribosomal switches to achieve better regulatory control of metals, and even to achieve specific selection of metals. We will explore different types of metal-binding proteins, based on the active binding site, and we are developing a computational model to predict the detection and concentration efficiency of synthetic biology biosensors for different metals.

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Isolation of cyanophycin from tobacco and potato plants with constitutive plastidic *cphA_{Te}* gene expression

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abstract

A chimeric cyanophycin synthetase gene composed of the *cphA_{Te}* coding region from the cyanobacterium *Thermosynechococcus elongatus* BP-1, the constitutive 35S promoter and the plastid targeting sequence of the integral photosystem II protein PsbY was transferred to the tobacco variety Petit Havana SRI and the commercial potato starch production variety Albatros. The resulting constitutive expression of cyanophycin synthetase leads to polymer contents in potato leaf chloroplasts of up to 35 mg/g dry weight and in tuber amyloplasts of up to 9 mg/g dry weight. Both transgenic tobacco and potato were used for the development of isolation methods applicable for large-scale extraction of the polymer. Two different procedures were developed which yielded polymer samples of 80 and 90% purity, respectively

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1. Introduction

Industrially relevant polymers cannot only be produced through chemical synthesis (Nawrath et al., 1995) but also by certain organisms. Microorganisms as well as plants are able to synthesize substances *e.g.* polyhydroxyalkanoates with properties similar to polyacrylates or other synthetic polymers, either naturally or after genetic modification (Börnke and Broer, 2010). A well known example is polyhydroxybutyrate (PHB), a polyester with properties similar to polypropylene (Anderson and Dawes, 1990; Poirier and Gruys, 2002) or the polydisperse biopolymer cyanophycin, which was described as a cell inclusion of cyanobacteria more than a hundred years ago (Borzi, 1887). In cyanobacteria, cyanophycin functions as a storage substance for energy and nitrogen (Simon, 1971).

Cyanophycin is formed in the cytoplasm and accumulates to a granular form (Allen, 1984). It is composed of a polyaspartic acid backbone to which arginine residues are bound. Arginine is linked *via* its α -amino functional group to the β -carboxyl group of each aspartate residue (Simon, 1976; Simon and Weathers, 1976). The skeleton of polyaspartic acid is of industrial interest alike the arginine residues. Polyaspartic acid can be used in biomedical (Leopold and Friend, 1995; Pratesi et al., 1985) as well as in numerous industrial and agricultural applications (Oppermann-Sanio and Steinbüchel, 2002; Schwamborn, 1998; Zott et al., 2001). It could replace the commonly used non-biodegradable polyacrylates and thus help to reduce environmental pollution. There is evidence which suggests that arginine, the second constituent amino acid of cyanophycin, has positive effects on humans in some immunological and physiological processes, for example as growth inductor (Lenis et al., 1999; Roth et al., 1995; Wu et al., 2007) or as a tumour cell inhibitor (Amber et al., 1988; Caso et al., 2004; Flynn et al., 2002).

Because of the low polymer content and their slow growth rate, cyanobacteria are inappropriate organisms for an industrial scale production of cyanophycin (Mooibroek et al., 2007; Sallam, 2009). By gene transfer cyanophycin synthesis was introduced into various heterotrophic microorganisms like *Escherichia coli*, *Ralstonia eutropha*, *Pseudomonas putida*, yeast, etc. that accumulate

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cyanophycin in their cells up to 50% of their dry weight (Aboulmagd et al., 2001; Diniz et al., 2006; Krehenbrink et al., 2002; Ziegler et al., 1998). Progress has also been made in the area of transforming the cyanophycin synthetase (*cphA*) gene into tobacco and potato plants (Hühns et al., 2008, 2009; Neumann et al., 2005). Tobacco is known to be a good system for the production of recombinant proteins (Salmon et al., 1998; Sehne et al., 1994; Wang et al., 2001), as it is easy to transform and shows great possibilities for scale-up due to the production of thousands of seeds per plant (Balasubramaniam et al., 2003). Cyanophycin accumulation up to 6.8% of dry weight (dw) in tobacco leaves and 7.5% dw in potato tubers have been reported (Hühns et al., 2008, 2009). However no large-scale isolation procedures that may be suitable for commercial applications have been established to isolate the polymer from plants.

In the present study, it is shown that the simultaneous, constitutive expression of cyanophycin synthetase is feasible in both chloroplasts and amyloplasts of potato plants. Furthermore, an isolation procedure for the polymer from cyanophycin producing transgenic tobacco and potato plants as well as a simple purity detection method was developed. For the first time a combination of the isolation process for the biopolymer cyanophycin to a commercial industrial process is discussed.

2. Materials and methods

2.1. Reagents

Hydrochloric acid for extraction was purchased from VWR (Leuven, Belgium). Sodium hydroxide was obtained from Lachema-Chemapol (Prague, Czech Republic) and sodium hydrogen sulphite was purchased from Merck-Schuchard (Hohenbrunn, Germany). Methanol and acetone were obtained from J.T. Baker (Deventer, USA) as 'Baker ultra resi-analysed' solution. The water used for the analysis was purified with a water purification system from SG Seral (München, Germany).

2.2. Sample preparation

For extraction of cyanophycin, freeze dried potato and tobacco leaf material as well as fresh potato tubers were used. Transgenic plants (potato: Albatros PsbY-*cphA*_{TE}; tobacco: PsbY-*cphA*_{TE}) were grown either in the field or in a greenhouse in 2006, 2007 and 2008. For lyophilisation, the tubers were cut into pieces, ground with a kitchen blender and freeze-dried (model ALPHA 1-4 LSC, Christ, Osterode, Germany). The dried potato samples were crushed with a coffee mill from Severin (Sundern, Germany) and passed through a 0.5 mm sieve. The fresh tubers were stored in the dark at a temperature of around 4 °C. Before extraction, fresh tubers were ground with a kitchen blender after addition of 0.1% (w/v) sodium hydrogen sulphite solution to prevent the enzymatic browning reaction.

2.3. Acidic extraction of cyanophycin

The acidic extraction was performed as described by Elbahloul et al. (2005). The plant material was extracted three to four times with 0.1 M hydrochloric acid. The solution was decanted and brought to pH 5 with 0.1 M sodium hydroxide for precipitation of cyanophycin. The aqueous solution was centrifuged (30 min; 10,000 g; 20 °C) and the precipitate was dried in a speed vac apparatus (Christ, Osterode, Germany). The yield of raw cyanophycin was determined gravimetrically.

2.4. Starch coupled isolation process

Fresh potato tubers were used for the extraction of cyanophycin in combination with the starch isolation process. The ground

potatoes were first stirred in an aqueous solution of 0.1% (w/v) sodium hydrogen sulphite for 1 h. Thereafter, the mixture of tuber pieces and aqueous solution was decanted through a sieve (0.5 mm pore diameter). The aqueous flow-through containing cyanophycin and starch granules were passed again through a sieve (70 µm pore diameter). The residue was treated with 0.1 M hydrochloric acid to dissolve the cyanophycin as described above. After separation from the insoluble starch, cyanophycin was precipitated by neutralisation with sodium hydroxide.

2.5. SDS-PAGE of cyanophycin

The dried cyanophycin samples were redissolved in 0.1% (w/v) sodium dodecylsulphate (SDS) for 1 h at 37 °C. 15 µL of this solution were mixed with 5 µL of 4 NuPAGE[®] LDS (lithium dodecyl sulphate) sample buffer from Invitrogen (Carlsbad, USA) and heated for 10 min at 70 °C. 10 µL samples were run per lane in a vertical mini-gel system from Sigma-Aldrich (single unit; Germany) using different Bis-Tris Novex[®] gels with a NuPAGE[®] running buffer at 70 V for at least 2 h. The standard protein samples for molecular weight comparison were purchased from Invitrogen (Novex[®] Sharp Unstained Protein Standard; Carlsbad, USA). Protein bands were stained using NuPAGE[®] Simply Blue solution from Invitrogen (Carlsbad, USA).

2.6. Analysis of water-soluble and water-insoluble forms of cyanophycin

Both forms of the polymer (Ziegler et al., 2002) were analysed according to the method described by Hühns et al. (2008).

Furthermore, a faster method for the quantification of cyanophycin from plant material was developed. This method is based on the protein assay developed by Bradford in 1976 which was first used by Obst et al. (2004) for the quantification of cyanophycin. Extraction of the water-insoluble cyanophycin form was performed with 0.1 M hydrochloric acid in several steps. For the measurements, the dye reagent concentrate from Biorad (München, Germany) was used as described in the manual. The hydrochloric acid used for extraction had to be diluted with water due to the intolerance of the dye reagent to high acid concentrations. A calibration curve was determined with cyanophycin isolated from *E. coli* (Ziegler et al., 1998) instead of bovine serum albumin, because the strong binding of the dye to aromatic amino acids and basic amino acids like arginine, which is abundant in cyanophycin, causes a much stronger colour development than the reaction with cell proteins. The differences between raw polymer concentrations, adjusted by dispersing a weighed aliquot of dried cyanophycin extract in a given volume of diluted acid and the concentrations determined by the Bradford assay were taken as impurity.

3. Results and discussion

3.1. Generation of transgenic *Solanum tuberosum* plants with plastidic production of cyanophycin

Although the formation of a biopolymer as a by-product in industrial starch potatoes seems to be a sustainable production system, first attempts to produce cyanophycin in the cytoplasm of potato cell was accompanied by strong changes in leaf morphology and decelerated growth with maximal cyanophycin contents up to 0.24% of dw (Neumann et al., 2005). Since the relocation of cyanophycin production into plastids of tobacco plants led to an increase in polymer accumulation up to 6.0% of dw without significant disturbance of plant growth and development (Hühns et al.,

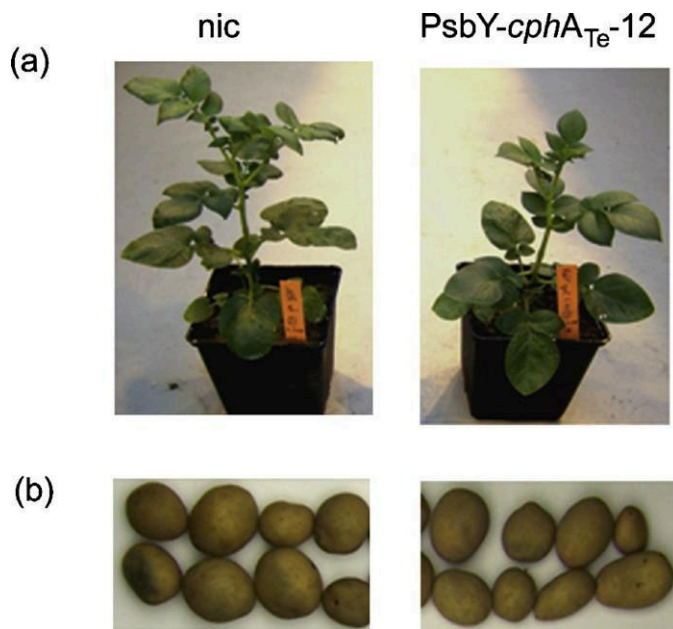


Fig. 1. Phenotype of a transgenic PsbY-cphA_{TE} potato plant in comparison with the near isogenic control plant (nic). (a) The PsbY-cphA_{TE} event 12 was indistinguishable in growth when compared to the near isogenic control. (b) Phenotype of tubers produced.

2008), the same strategy was used to produce cyanophycin in amyloplasts and chloroplasts of potato.

In order to induce production of cyanophycin in the plastids of potato, the coding region of the cyanophycin synthetase gene isolated from *Thermosynechococcus elongatus* BP-1 was fused to the PsbY translocation pathway signal peptide for import into chloroplasts (Hühns et al., 2008). The transit peptide belongs to PsbY, an integral protein of photosystem II (Gau et al., 1998), which has previously been shown to mediate the import of cyanophycin synthetase into tobacco chloroplasts (Hühns et al., 2008) and potato amyloplasts (Hühns et al., 2009). The construct, named pPsbY-cphA_{TE}, contained the *nptII* gene as selectable marker (Herrera-Estrella et al., 1983) and was used for five transformation experiments with *S. tuberosum* cultivar 'Albatros' (Table 1). The obtained transgenic plants expressing cyanophycin synthetase were named PsbY-cphA_{TE}.

In comparison to plants transformed with the control vector pLH9000, carrying only the antibiotic resistance gene *nptII* and the p35S promoter without the *cphA_{TE}* coding region, reduced regeneration frequencies were observed for plants transformed with pPsbY-cphA_{TE} (Table 1). Nonetheless, growth and appearance of the shoots of plants transformed with PsbY-cphA_{TE} were indistinguishable from that of control plants (Fig. 1a). The presence of the *cphA_{TE}* transgene in potato plants was proven by PCR analysis of leaf extracts. The majority of generated plants contained the *cphA_{TE}* gene (data not shown), and the presence of the transgene specific full length mRNA of *cphA_{TE}* was demonstrated by Northern blot analysis in 9 of 14 of the regenerated clones (Table 1). Compared to both, the near isogenic control and LH9000 plants, all transgenic PsbY-cphA_{TE} potato clones analysed exhibited no significant phenotypical changes or growth retardation in *in vitro* as well as in greenhouse cultivation experiments. Tubers were produced by 9 of the 9 analysed PsbY-cphA_{TE} plants, and showed normal size (Fig. 1b).

Unlike cytosolic expression, the constitutive plastidic expression led to normal tuber weight with no significant changes in phenotype compared to the near isogenic control (Fig. 1b). The constitutive plastidic expression of cyanophycin synthetase leads

to a 1.6-fold increase of its accumulation in leaves and a 9.0-fold increase of the polymer content in tubers compared to the constitutive cytosolic expression under control of the CaMV35S promoter (Neumann et al., 2005). This increased polymer accumulation without deleterious effects on the plants might be possible because plastids, which can be considered as descendants of cyanobacteria, may provide a more appropriate environment than the cytoplasm of plant cells. The key enzymes that are required for biosynthesis of the amino acid constituents of cyanophycin L-aspartate and L-arginine are present in the chloroplast (Chen et al., 2006; Forchhammer and Demarsac, 1994; Slocum, 2005; Sugiyama et al., 2004).

Apparently, the transgenic plants favour the cyanophycin accumulation in chloroplasts according to the amounts detected in leaves and tubers. Generally, in chloroplasts the reduction of nitrate for amino acid production is a very energy consuming process and is linked to the photosynthesis. In addition to the normal demand of the plant, the cyanophycin production displays a new consumer of metabolites like amino acids. An important benefit of the cyanophycin production in chloroplasts in comparison to cytoplasm may be that the produced amino acids aspartate and arginine could be directly used for polymer accumulation. Foreign protein accumulation was already shown to be lower in amyloplasts than in chloroplasts (Nguyen et al., 2005; Sidorov et al., 1999).

3.2. Analysis of cyanophycin content and electron microscopic examination of transgenic potato plants

The cyanophycin content in leaves and tubers of all investigated PsbY-cphA_{TE} clones was measured twice. Cyanophycin was found in leaves and tubers of each of the 9 PsbY-cphA_{TE} clones (Table 2). Transgenic plants primarily produced the water-insoluble form of the polymer and only small amounts of the soluble form (Table 2). The mean cyanophycin content in the leaves of all analysed potato clones was 16.0 mg/g in dw, with a maximum amount of 37.0 mg/g dw in clone 12 (Table 2) whereas in tubers, the mean content was 5.6 mg/g dw, with a maximum amount of 9.8 mg/g dw in clone 12 (Table 2).

In electron micrographs of PsbY-cphA_{TE} plant leaves and tubers, cyanophycin aggregates were exclusively visible in chloroplasts (leaves) and amyloplasts (tubers) (Fig. 2a and b). Like in tobacco and potato after constitutive or tuber specific expression, respectively (Hühns et al., 2008, 2009; Neumann et al., 2005), the aggregates were composed of cyanophycin strands, which differed in thickness and length. These were arranged in piles, and reacted with an anti-serum raised against cyanobacterial cyanophycin (Fig. 2c) and also with an antiserum raised against the cyanobacterial cyanophycin synthetase (data not shown). Fortunately, neither leaves nor tubers of the transgenic potatoes were affected by the cyanophycin production. In contrast to that, the tuber-specific plastidic expression of the transgene under control of the B33 promoter led to stress symptoms like brown spots starting at tubers navels. Since these tubers contain high cyanophycin rates of up to 7.5% dw, it might be speculated that the unaffected phenotype of the tubers described here is due to the relatively low amount of cyanophycin. Nevertheless, the capability to cope with high cyanophycin concentrations might also be species or tissue specific since tobacco leaves were able to produce up to 6% of dw without any sign of stress (Hühns et al., 2008).

A water soluble form of cyanophycin is formed in tobacco and potato as a second polymer type. The ratio of these two forms differed in PsbY-cphA_{TE} plants. In transgenic PsbY-cphA_{TE} tobacco plants producing cyanophycin in the chloroplasts of leaves, the mean ratio of insoluble to water-soluble form was 1:0.003 (Hühns et al., 2008). In leaves of PsbY-cphA_{TE} potato plants a 10-fold

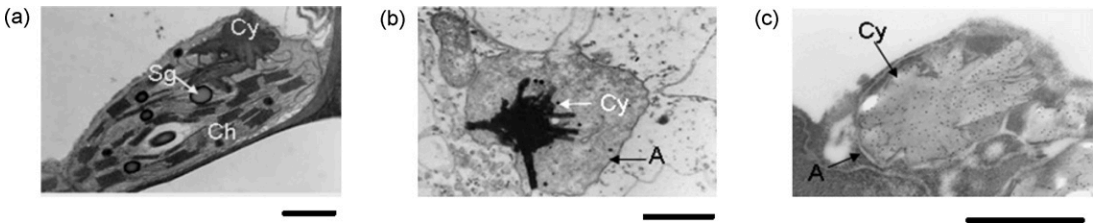
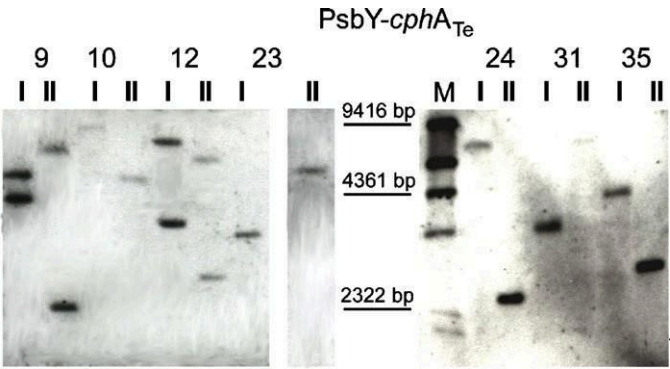


Fig. 2. Electron microscopy of cyanophycin in leaves and tubers of PsbY-*cphA*_{T_e} potato plants: (a) in leaves and (b) in tubers. (c) Immunocytochemical detection of cyanophycin by the anti-cyanophycin-antiserum and gold-coupled anti-rabbit immunoglobulin G (IgG) antibody (Bars, 1 μm; Cy, cyanophycin; Ch, chloroplast; Sg, starch grain; A, amyloplast).

increased ratio of the water-soluble form was observed (1:0.04). In transgenic PsbY-*cphA*_{T_e} potato tubers a similar mean ratio of 1:0.05 was found. Amazingly, the chemical structure of the two forms of the polymer has been reported to be identical (Füser and Steinbüchel, 2005). The soluble form of cyanophycin was also found in the recombinant *Saccharomyces cerevisiae* expressing *cphA* (Steinle et al., 2008) and occurred naturally in the non- cyanobacterial eubacterium *Desulfitobacterium hafniense* (Ziegler et al., 2002).

3.3. Determination of the transgene copy number and transgene specific RNA accumulation in PsbY-*cphA*_{T_e} potato clones

In order to investigate the influence of T-DNA integration and copy number of the *cphA* gene in PsbY-*cphA*_{T_e} clones on cyanophycin production, genomic DNA from leaves of PsbY-*cphA*_{T_e} clones 9, 10, 12, 23, 24, 31 and 35 were evaluated by Southern blot hybridisation (Fig. 3). DNA was digested with *Bcl*II and *Eco*RI, blot- ted and hybridised with the *cphA* probe. For PsbY-*cphA*_{T_e} clones 9 and 12, two bands were observed for *Bcl*II- and *Eco*RI-digested DNA probed with the *cphA* gene. This analysis indicated that two copies of the *cphA* gene had been integrated into different loci of the genome. In contrast, a single band was observed for PsbY-*cphA*_{T_e} clones 10, 23, 24, 31 and 35, which indicates a single locus T-DNA insertion event (Fig. 3).



of DNA digested with *Bcl*II or *Eco*RI was probed with the *cphA*_{T_e} coding sequence (M, marker). The sizes of the DNA fragments in base pairs (bp) are indicated between the blots. Lane I, DNA digested with *Bcl*II; lane II, DNA digested with *Eco*RI. At the bottom transgenic events are indicated.

3.4. Use of an acidic extraction procedure for cyanophycin isolation from plants

The experiments were done first with freeze dried plant material, because the material has to be stored over a long period of time and normally fresh plant material would undergo special degradation processes and therefore be unusable for additional

Table 1
Summary of the regeneration rates and characterisation of regenerated transgenic plants. *Solanum tuberosum* transformed with plastidic expression of the cyanophycin synthetase, in comparison to the control vector pLH9000.

Regenerated transgenic plants		Plants expressing <i>cphA</i> _{T_e} mRNA		Phenotypical changes
Construct	Shoot/ xplant (mean rate)			

P
l
a
n
t
s

p
r
o
d
u
c
i
n
g
c
y
a
n
o
p
h
y
c
i
n
i
n
t
u
b
e
r
s

					Of leaves	Of tubers
pLH9000	0.46	5	0 of 5	0 of 5	0 of 5	0 of 5
pPsbY- <i>cphA</i> _{Te}	0.27	14	9 of 14	0 of 9	0 of 9	9 of 9

Table 2
Determination of cyanophycin content and composition (soluble and insoluble forms) in tubers and phenotype of various transgenic potato tubers (n.d., not detected).

Plant	Transgenic event	Content of cyanophycin in leaves (mg/ g dry weight)			Content of cyanophycin in tubers (mg/ g dry weight)			Weight of tubers (g)
		Total	Insoluble	Soluble	Total	Insoluble	Soluble	
LH9000	53	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	69.69
PsbY- <i>cphA</i> _{Te}	9	23.2	21.9	1.3	7.4	7.2	0.2	59.89
	10	29.2	28.3	0.9	8.3	8.0	0.3	38.42
	12	37.0	35.3	1.7	9.8	9.3	0.5	47.04
	21	0.5	0.4	0.1	6.1	5.8	0.3	28.38
	23	10.3	9.5	0.8	9.3	8.9	0.4	51.93
	24	21.1	20.5	0.6	5.0	4.7	0.3	50.33
	25	0.2	0.2	n.d.	0.2	0.1	0.1	35.85
	31	11.5	10.8	0.7	2.4	2.3	0.1	9.09
	35	11.0	11.0	n.d.	2.1	2.0	0.1	21.54

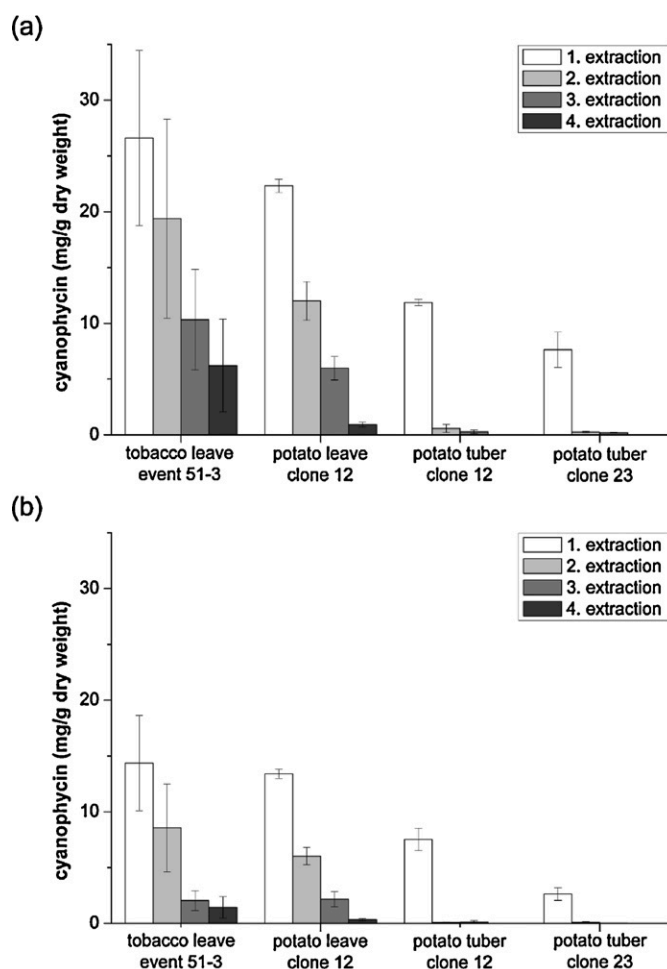


Fig. 4. Yields of cyanophycin extraction from transgenic plant material obtained by consecutive extractions: tobacco, *PsbY-cphA_{TE}* event 51-3; potato, *Albatros PsbY-cphA_{TE}* clones 12 and 23 ($n = 3$). (a) Yields without correction for purity; (b) yields corrected for purity of the isolated cyanophycin samples (purity was determined by the Bradford method).

experiments. On the other hand all results could be correlated to the dry weight of the used plant material and in this way were easy to compare.

The first goal of our work was to investigate the minimum number of acidic extractions for sufficient cyanophycin yields. With four extractions of the freeze-dried *PsbY-cphA_{TE}* tobacco leaves around 80% of the total extractable amount of cyanophycin could be isolated. For the freeze-dried potato leaves tested (*PsbY-cphA_{TE}* clone 12 and 23) three extraction steps are sufficient to yield over 90% of the total amount of cyanophycin (Fig. 4a). The extraction of freeze-dried potato tuber material resulted in the recovery of over 90% of the total extractable cyanophycin after just one extraction. The amount of cyanophycin was determined after extraction and precipitation from acidic solution with sodium hydroxide by weighing the dried pellet. A total of 75.7 mg/g dw of cyanophycin could be extracted from the tobacco leave material and 41.2 mg/g dw of cyanophycin resulted from the extraction of the potato leaves.

Fig. 4b illustrates yields of cyanophycin corrected for the purity of isolated samples. Especially polymer samples from tobacco leave gave only less purity. The purity decreases from 54% in the first extraction step to 23% in the last step. For cyanophycin samples from potato leave the purity was in all extraction steps higher than for the ones from tobacco. The first extract has a purity of 60% whereas the fourth step leads to samples with a decreased purity

of 35%. The extraction of lyophilized potato tuber material of clone 12 leads to polymer samples with 65% purity. Whereas isolated cyanophycin samples of tuber material of clone 23 was only pure with 35%. Samples of the second and third extraction steps the tuber material can be omitted, because of the less extraction yield.

The next experiments were performed to determine the efficiency of various acids for the extraction of the biopolymer from freeze-dried plant material. For the measurements described above cyanophycin was extracted with hydrochloric acid. However, this acid may lead to corrosion of the steel equipment commonly used in industrial processes. Thus sulphuric acid and formic acid were also used to isolate the biopolymer. For the extraction, every acid solution needs to have a pH-level less than two. Hence, the concentrations of the acids were chosen to match this requirement. The results are shown in Table 3. Best results for the extraction were achieved using hydrochloric acid, with efficiency three to five times greater than with the other acids investigated. One explanation for the different extraction yields could be the acidity constant. Hydrochloric acid ($pK_a = -6.0$) is the strongest of the acids tested in this work, followed by sulphuric acid ($pK_a = -3.0$) and formic acid ($pK_a = 3.75$) (Rauscher et al., 2000; Riedel and Janiak, 2007). Among the commonly used acids, only perchloric acid is more acidic than hydrochloric acid and could perhaps yield in higher amounts of biopolymer per extraction step. However due to the corrosive impact of this acid to common steel equipment the application at industrial scale is not advantageous either. Thus hydrochloric acid was used in the experiments described below.

The next step of cyanophycin isolation which had to be optimized was the separation of the polymer from the acidic solution by pH adjustment with sodium hydroxide. The solubility of zwitterions such as proteins reaches a minimum at their isoelectric point (pI). At a pH corresponding to the pI , the protein has no net charge. Its recovery would be lower at pH values near the pI . Studies of the pI of cyanophycin from cyanobacteria showed that there are two forms with slightly different isoelectric points, one at pH 4.75 (over 97% of cyanophycin) and the other at pH 6.10 (Simon and Weathers, 1976). Several experiments were carried out to determine the best pH for the precipitation of cyanophycin. Therefore the acidic solution was set with sodium hydroxide to pH-levels ranging from 3 to 7.

The highest yields of cyanophycin precipitation were achieved around pH 5 (Fig. 5c). Thus, in the isolation procedure described below the pH of the solution used for extraction of cyanophycin from the plant material was adjusted to 1 (0.1 M HCl), whereas pH 5 was used for separation of cyanophycin from the acidic extraction solution.

3.5. Determination of cyanophycin purity

The cyanophycin amount in the transgenic plant material was measured with an enzymatic test system described by Ziegler et al. (2002) and Hühns et al. (2008). Discrepancies of up to 50% were detected between the results obtained by the enzymatic test and the masses of dried isolates from the acidic extraction procedure (data not shown). It was therefore necessary to determine the purity of the isolated cyanophycin samples.

Sodium dodecyl sulphate-polyacrylamide gel electrophoresis (SDS-PAGE) was used for determination of impurities from proteins since the acidic extraction method may not only extract cyanophycin but also other substances such as acid soluble proteins. The average pI of potato proteins lies near pH = 5 (Lotz and Eggengoor, 2008) and also tobacco has more acidic proteins ($pI < 7$) than basic ones ($pI > 7$) (Balasubramaniam et al., 2003). Thus, these proteins may be extracted and precipitated by the acidic extraction procedure due to their isoelectric points.

Table 3Cyanophycin yield after extraction from fresh transgenic potato tubers (PsbY-*cphA*Te clone 23) with different acids.

Acid	pKa	Acid (mol/L)	Isolate (g)	Isolate (mg/g dry weight)
Formic acid	3.8	0.6	52.8	1.6
Sulphuric acid	~3.0	0.01	24.1	0.7
Sulphuric acid	~3.0	0.1	86.5	2.7
Hydrochloric acid	~6.0	0.1	204.7	6.5

For example patatin, the major storage protein from potato tubers has a *pI* value between 4.2 and 5.2. The wide range has its origin in the different isoforms of the protein (Racusen and Foote, 1980). The bands at 44 and 88 kDa could represent patatin and its dimeric form (Fig. 5a). When isogenic potato tubers were subjected to the acidic cyanophycin isolation process as a control, the same protein bands were observed (data not shown). The size range of the cyanophycin isolated from the transgenic plants was very different from that reported for cyanobacterial cyanophycin. The apparent molecular mass of the plant-derived polymer ranged just from 17 to 24 kDa (Fig. 5a and b), whereas in cyanobacteria cyanophycin can

be as large as 125 kDa (Simon, 1971). This appears to be a feature usually found with heterologously produced cyanophycin (Sallam, 2009). For identification of the cyanophycin bands in SDS-PAGE gels, the samples were incubated with Cyanophycinase CphB, a specific protease from cyanobacteria (Richter et al., 1999).

The highest amount of protein impurities precipitated when the acidic extracts were brought to pH 5 or 6, respectively, *i.e.* the pH values which were most efficient in precipitating cyanophycin from the plant extracts (Fig. 5c). By using the protein determination method of Bradford (1976) which can also be used to quantify cyanophycin (Obst et al., 2004), we checked the quantitative purity

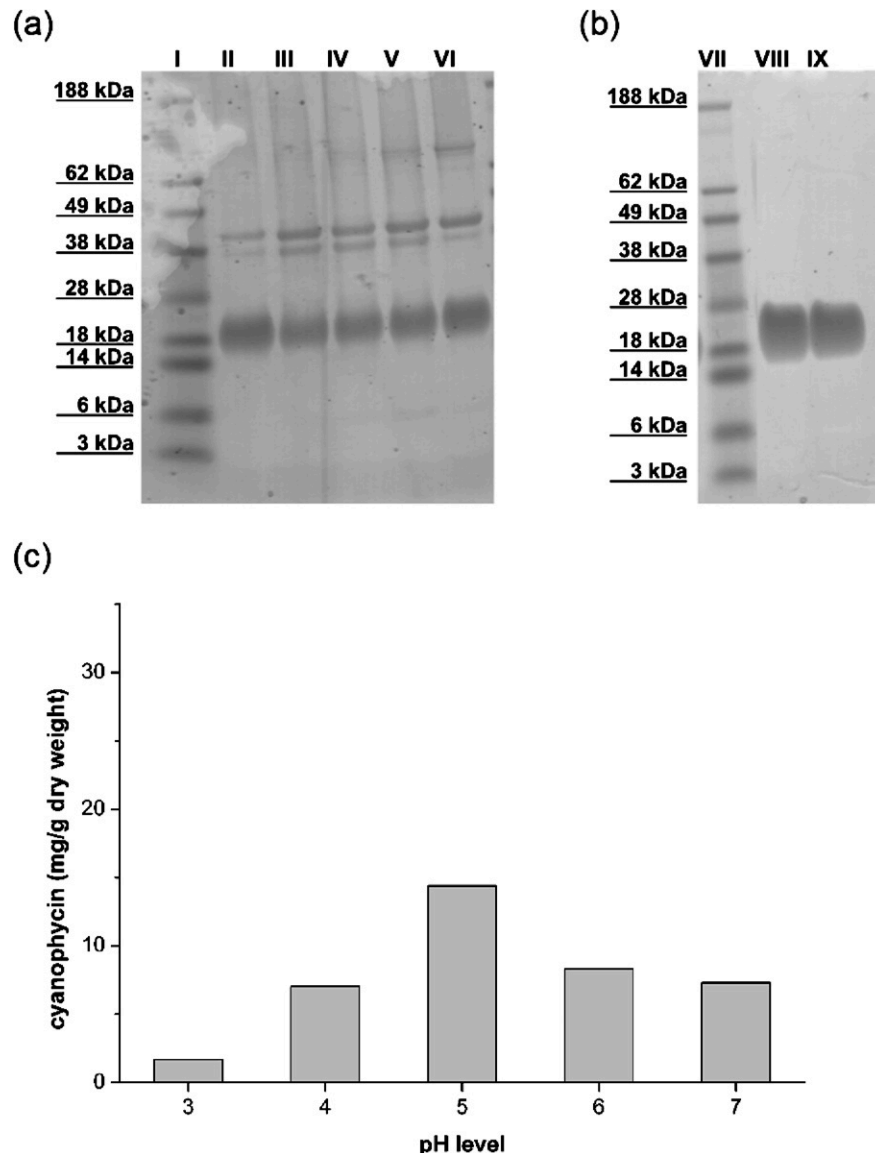


Fig. 5. (a) SDS-PAGE analysis of isolated cyanophycin samples. Lane I: molecular weight marker (kDa); lanes II–VI: cyanophycin precipitated at pH values ranging from 3 to 7. (b) SDS-PAGE analysis of isolated cyanophycin samples. Lane VII: molecular weight marker (kDa); lanes VIII and IX: cyanophycin from potato tubers isolated by the here described starch-coupled isolation process (PsbY-*cphA*Te clone 12 and PsbY-*cphA*Te clone 23). (c) The diagram shows the amounts of cyanophycin obtained by the acidic extraction method with HCl of fresh transgenic potato tubers (PsbY-*cphA*Te clone 12) at different pH levels (experiment was done only once).

Table 4 Efficiency of laboratory starch isolation and coupled cyanophycin extraction in comparison of tubers from different transgenic potato clones with near isogenic control tubers (n.d., not detected; $n = 3$).

Plant	Content in plants		Extracted substance	
	Starch (% fresh matter)	Cyanophycin (% dry weight)	Starch (% fresh matter)	Cyanophycin (% dry weight)
Near isogenic control (nic)	19.4 ± 1.9	n.d.	11.8 ± 4.1	<0.1
PsbY-cphA1e clone 10	n.d.	0.8 ± 0.3	8.2 ± 0.8	0.6 ± 0.2
PsbY-cphA1e clone 12	17.4 ± 0.6	0.9 ± 0.6	8.5 ± 2.1	0.7 ± 0.2
PsbY-cphA1e clone 23	19.2 ± 1.2	1.3 ± 0.2	7.3 ± 0.0 ^a	0.4 ± 0.0 ^a

^a Experiment was done only once.

of every sample. Due to the strong affinity of the Bradford reagent (Brilliant Blue G-250) to basic amino acids like arginine and to aromatic amino acids, the commonly used calibration standard bovine serum albumin (BSA) was replaced by pure water insoluble cyanophycin isolated from recombinant *E. coli* (Ziegler et al., 1998) for the determination of cyanophycin concentrations. The slope of the calibration curve is three times higher than that of the standard calibration with BSA (data not shown). In this way the Bradford test could be used for checking the purity of cyanophycin samples. For polymer samples from potato tuber material precipitated at pH 3 and 4, respectively, the purity nearly reached 50%, whereas for the polymer isolated at pH values of 5, 6 and 7, the purity decreased to about 40 to 30% (data not shown).

The impurities could be identified only in part so far. About 13% of the impurities were estimated to be due to potato proteins co-extracted by the acidic extraction method. The remaining impurities could be soluble forms of starch and metabolites of the potato tubers, or salt residues from the isolation process. With respect to further treatments of the biopolymer like partial hydrolysis of the isopeptide bonds, which connect the polyaspartate backbone with the arginine residues, to yield polyaspartic acid and free arginine, one has to consider whether the impurities could either enhance or affect these processes. Thus it might be necessary to develop further purification steps or modifications of the above described method to increase the purity of cyanophycin as described below.

3.6. Optimisation of cyanophycin purity by a starch-coupled isolation process

An alternative extraction procedure for cyanophycin from starch-containing potato tubers was developed. The method is based on a well-established starch isolation process which takes advantage of the special solubility behaviour of the polysaccharide in aqueous solutions. The isolation of cyanophycin close to the starch extraction could result in a value enhancement for the starch industry.

The first step in the starch isolation process is the disruption of the tubers and the suspension in large volumes of water to allow the separation of starch from cellular debris and soluble materials (Knorr et al., 1977). Thereafter the starch granules are sedimented and dried. At this point it should be possible to combine the isolation procedure for starch with the acidic extraction method for the isolation of cyanophycin (Fig. 6). Because of the insolubility of most of the polymer at neutral pH (Table 2) and of its density, cyanophycin can be separated from the soluble potato material together with the starch granules. The precipitate of starch and cyanophycin can then be extracted with acid as described above, and thus the purity of polymer increased due to the separation of soluble proteins and other soluble contaminants in the previous step.

Several experiments were performed to verify this method. In contrast to the cyanophycin samples from the acidic extraction

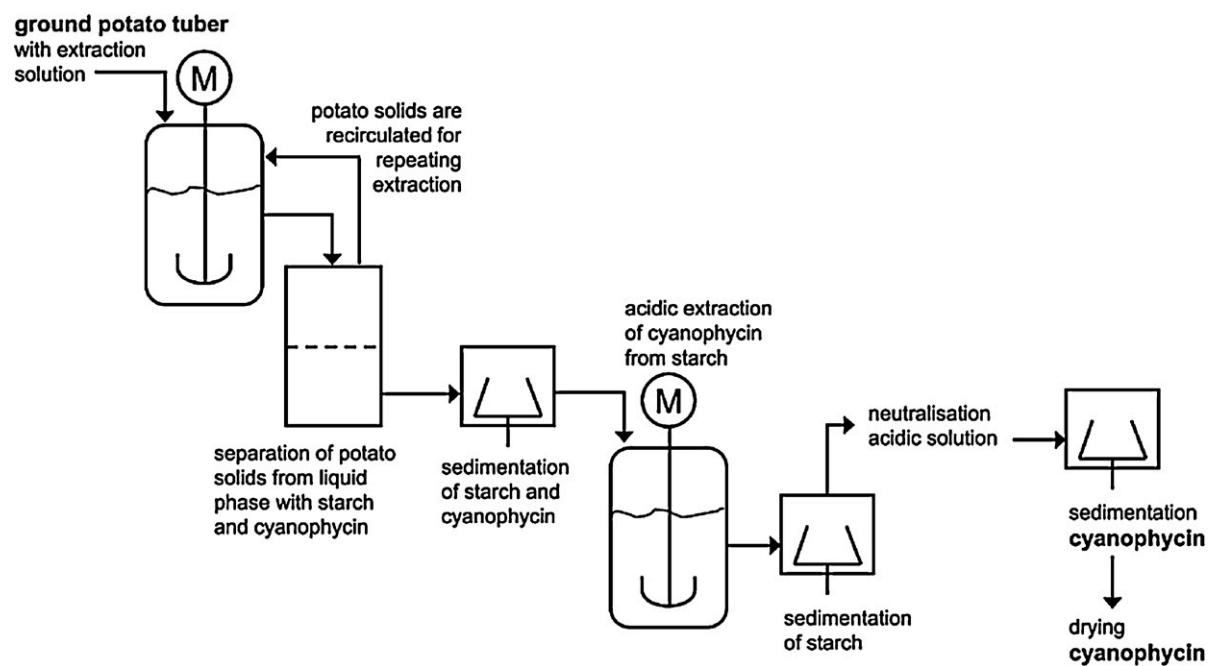


Fig. 6. Schematic representation of the isolation process for starch from potato tubers combined with acidic extraction method for cyanophycin ("starch-coupled isolation process").

method (Fig. 5a, lines II–VI), SDS-PAGE analyses of cyanophycin samples obtained by the starch-coupled isolation process showed no protein contaminations (Fig. 5b, lines VIII and IX). Analysis of various dried and weighed cyanophycin samples with the Bradford method (see above) suggests a purity of cyanophycin ranging over 80–90%. Some of the analysed polymer samples contained over 90% cyanophycin. The remaining impurities were not yet identified, but one possibility is that salt residues from the isolation procedure adhere to cyanophycin. Up to now, the required purity for industrial applications is unclear because no applications for the direct usage of cyanophycin have been developed. However, the polyanionic backbone of cyanophycin is of interest to the industry, as well as its other constituent arginine, which can be used e.g. as dietary supplement (Sallam, 2009).

All in all the isolation of cyanophycin from industrially used potatoes would result in a value enhancement for the whole process. This may positively influence the price for the product cyanophycin. A further increase of the amount of cyanophycin in transgenic plants will also reduce the costs for the isolated polymer. This is one goal of future research.

Furthermore the efficiency of the starch-coupled isolation method at laboratory scale was determined. In Table 4 we compare the starch and cyanophycin levels in potato tubers prior to the experiments with the extraction values after isolation. First of all one can recognize that starch isolation is not efficient at laboratory scale. Only up to 60% of starch could be recovered, whereas the cyanophycin values prior to and after extraction of the polymer show good agreement. Although the starch isolation in the laboratory is not a perfect solution the isolation of the polymer is nearly 100%. We could not detect any changes in the percentage nor the quality of isolated starch by additional acidic washing steps (data not shown).

The starch-coupled isolation process discussed in the paper was also transferred to the potato plants which contain, due to the tuber-specific plastidic cyanophycin expression (pB33-PsbY-cphAte), drastically higher amounts of cyanophycin (Hühns et al., 2009). These tubers can accumulate the biopolymer up to 7.5% of dry weight. Higher polymer contents in tubers lead to accordingly higher extraction values of cyanophycin and therefore make the whole production process in plants more competitive.

4. Conclusion

In this study, the isolation of polymer cyanophycin from transgenic tobacco and potato plants with plastidic expression of cyanophycin synthetase was demonstrated for the first time. A commercialisation of the isolation process depends on a method which is both efficient and cost-effective. Hence, the idea was to combine a well-established industrial process with the isolation of the biopolymer and/or to modify the genetic structure of plants in a way that would allow several applications.

Compared to bacterial production of biopolymers, one advantage of using plants is the reduction of costs for production (low costs for energy and substrates). However, bacterial or fungal fermentation cycles are usually much shorter than the growth periods of plants in the field. This can only be compensated by high accumulation rates in the plants, large areas under cultivation and low production costs. In addition, cheap isolation methods that are attached to the isolation of the primary product like starch will increase the competitiveness of plant produced cyanophycin.

Our new starch-coupled isolation method for the biopolymer cyanophycin from potato tubers was also applied for the already described plants with tuber-specific plastidic cyanophycin expression (pB33-PsbY-cphAte) (Hühns et al., 2009). Again, 80–90% of the cyanophycin could be isolated, which increased the overall

cyanophycin yield due to the higher polymer content in the plants. Therefore, the efficiency of the method does not depend on the cyanophycin content in different transgenic potato lines.

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